UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Physics

NED University of Engineering and Technology

Radiation Dosimetry of Worker during Different Dental Imaging Procedure

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Author's Declaration

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Statement of Contribution

Project Overview:

Our group collaborated on a project focused on radiation dosimetry in dental imaging procedure. The aim was to ensure the safety of patients, healthcare professionals, and the public by assessing the effectiveness of the radiation shielding in protecting against harmful radiation exposure.

Ayesha bint e Arif:

Ayesha, as the project lead, played a pivotal role in team organization and ensuring seamless communication among all members. Her responsibilities encompassed overall project management and coordination. Additionally, she actively contributed to coordinating the writing of the thesis, with a specific focus on formatting the document. She also did the calculation of linear regression.

Dua Arif:

She plays an important role in conduction of readings and calculations. She was coordinating the writing of the thesis, specifically crafting the abstract and summarizing the project's objectives. Additionally, she was responsible for typing the calculations of dose measurement and the regression analysis.

Hiba Azam:

Hiba assumed the responsibility of meticulously documenting the project's progress and organizing all research findings into a cohesive and well-structured thesis. She plays major role in making the presentation.

Anas Nawaz:

He significantly took part in conducting the reading of dose rate, actively setting the angle and distance mark, phantom position. He has an active role in presentation and gathering the thesis result.

Executive Summary

This project investigates the issue of occupational radiation exposure to dental professionals during various imaging procedures, emphasizing the importance of safety in clinical environments. With the increased use of X-ray technology in dentistry, monitoring and minimizing exposure has become essential for protecting healthcare workers.

The report begins with an overview of radiation physics, including the nature of X-rays, their interaction with matter, and associated health risks. While dental X-rays are generally considered low-dose, repeated exposure can result in cumulative effects if appropriate precautions are not taken. Therefore, the implementation of safety strategies such as shielding, proper equipment use, and adherence to guidelines is critical.

To assess exposure, the study employed the Best X-CD intraoral X-ray device and FS2011+ personal dosimeter. These tools enabled real-time monitoring of radiation doses during clinical scenarios. The dosimeter was calibrated for accuracy, and controlled experiments were conducted by varying tube voltage (kVp), current (mA), exposure time, distance from the source, and angular orientation.

Findings revealed that radiation dose levels are significantly influenced by these parameters. Higher voltage and closer distances resulted in increased exposure, while different angles affected the distribution of scattered radiation. The FS2011+ dosimeter proved reliable in providing continuous, real-time data, allowing for informed adjustments during procedures. Statistical analysis, including linear and quadratic regression models, demonstrated strong predictive capabilities, enabling accurate estimation of dose levels in similar operational settings.

The results confirm that radiation exposure during dental imaging can be effectively managed and maintained within the occupational limits recommended by international regulatory bodies. With proper calibration, use of dosimetric tools, and consistent application of safety measures such as the use of lead aprons, optimized positioning, and adequate shielding healthcare workers can remain protected from harmful radiation effects.

The study reinforces the necessity of integrating dosimetry into daily dental practice. It also highlights the importance of regular training, strict compliance with radiation safety protocols, and continuous monitoring to safeguard the health of dental staff. This research contributes to the advancement of safer radiographic practices in dentistry and supports efforts toward improving occupational health standards in clinical imaging settings.

Acknowledgments

We would like to express our deepest gratitude and appreciation to all those who have contributed to the completion of this thesis. Their support, guidance, and encouragement have been invaluable throughout this academic journey. First and foremost, we would like to express our sincere appreciation to our project Supervisor, [Sir Faizan] , for his exceptional guidance, unwavering support, and valuable insights throughout the duration of this project. Their expertise and mentorship have been crucial in shaping the direction of our research. We would also like to thank our project Co. supervisor, [Sir Mujeeb ur Rehman], for his valuable input and constructive criticism, which helped refine this thesis. We are deeply thankful to [Dental Department, Dow university Of Health And Science] for granting us access to their X-dc, OPG machines. Their cooperation and assistance were essential in acquiring the necessary data for our study. We would also like to acknowledge the [NED University of Engineering & Technology] for providing the necessary resources, library facilities, and a conductive academic environment, which greatly facilitated our research. Although every effort has been made to ensure the accuracy and reliability of our findings, we acknowledge that there may be limitations in our research. The complexities of radiation shielding design and evaluation present challenges and we recognize the potential for further exploration in this field. In conclusion, we are deeply grateful to all those who have contributed to the successful completion of this project. Your support and encouragement have been indispensable, and we appreciate each and every one of you for being part of this remarkable journey.

United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace and justice. There is a total of 17 SDGs as mentioned below. Check the appropriate SDGs related to the project.

- ☐ No Poverty
- ☐ Zero Hunger
- ✓ Good Health and Well being
- ☐ Quality Education
- ☐ Gender Equality
- ☐ Clean Water and Sanitation
- ☐ Affordable and Clean Energy
- ☐ Decent Work and Economic Growth
- ☐ Industry, Innovation and Infrastructure
- ☐ Reduced Inequalities
- ☐ Sustainable Cities and Communities
- ☐ Responsible Consumption and Production
- ☐ Climate Action
- ☐ Life Below Water
- ☐ Life on Land
- ☐ Peace and Justice and Strong Institutions
- ☐ Partnerships to Achieve the Goals

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4	[Anas Nawaz]	[PH-21068]

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Sir Faizan Ahmed

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Chapter One

Introduction

1.1 X-Rays

1.1.1 What are X-rays?

In 1985, German engineer Wilhelm Röntgen made an extraordinary discovery while studying cathode rays in a gas discharge tube. He identified a unique kind of radiation capable of passing through opaque materials, causing fluorescence, and ionizing gases. Röntgen named this new radiation X-rays due to its high energy and short wavelengths.

Since then, X-rays have found wide-ranging applications in various fields such as medicine, industry, and research. In the field of medicine, X-rays play a crucial role in imaging the internal structures of the human body, aiding in the diagnosis and monitoring of diverse medical conditions. In the industries, within the realm of research, X-rays are utilized to delve into the atomic and molecular structures of various substances, enabling scientists to gain insights into their composition and behavioral sector, X-rays are employed for non-destructive testing, allowing for the inspection of internal structures and the identification of defects like cracks or voids in different materials.

1.1.2 X-Ray Tube

The tube comprises a glass envelope that is evacuated to a high vacuum and consists of two primary components: the Cathode (the filament that produces the electron beam) and the Anode (the target). The tube is additionally shielded to minimize the escape of X-rays and safeguard the operator from exposure to X-rays. The energy of X-rays can be adjusted by altering the voltage and current supplied to the cathode. The X-ray tube is housed in a vacuum chamber to prevent electrons from combining with air molecules; as such interactions would result in energy loss.

1. **The Anode**

The choice of tungsten for anode and cathode (target and filament) relies on its high atomic number, specifically Z=74, and high melting point, around 3370°C, that is necessary to enhance X-ray production energy. The copper anode serves to carry excess heat away to avoid causing damage to the target. Rotating anodes have been utilized in diagnostic X-rays to reduce the temperature of the target, everywhere. Certain stationary anodes are covered with a copper and tungsten shield to stop stray electrons from hitting the non-target parts of the tube. The X-ray emission point needs to be minimized in size to achieve clear radiographic images.

2. **The Cathode**

The cathode in an X-ray tube acts as the electron source for generating X-rays. It is made up of a tungsten filament that is heated through an electric current, causing thermionic emission and the release of electrons. Tungsten is used as a filament because it has very high melting point that enables it to withstand the required high temperatures. The low work function of tungsten will facilitate easy electron emission. The electrons emitted from the cathode are then accelerated towards the anode using a high-voltage power supply, creating an electric field. The cathode is typically positioned at the center of the X-ray tube, surrounded by a focusing cup that aids in directing the emitted electrons onto the anode.

1.1.3 Types of X-Rays

There are two types of X-rays.

1. Bremsstrahlung X-rays

Bremsstrahlung X-rays, also known as braking radiation or deceleration radiation, are a type of X-rays produced when high-energy electrons quickly slow down while passing through materials. A rapidly moving electron will bend and lose part of its kinetic energy upon interacting with positively charged atomic nuclei in a material. X-rays are emitted due to this energy depletion. With a minimal amount of energy required to surpass the atomic electron's binding energy, the energy from the emitted X-rays matches the energy lost by the electron.

Bremsstrahlung X-rays can be produced in various situations, including X-ray tubes, nuclear reactions, and particle accelerators. Bremsstrahlung X-rays exhibit a continuous spectrum with energy levels that can vary from zero up to the maximum energy of the incoming electrons. This suggests that the X-rays possess multiple energies instead of merely a single one.

. **2. Characteristic X-rays**

Characteristic X-rays, referred to as fluorescent X-rays, are a form of X-ray radiation released when high-energy electrons displace inner-shell electrons from atoms, resulting in an ionized atom in an excited condition. The energized atom subsequently emits energy as X-rays to revert to its ground state.

The X-rays generated during this process possess a distinctive energy that is unique to the element responsible for their creation, which makes them valuable for determining the makeup of a material. The energy of the characteristic X-rays corresponds to the energy difference between the atom's initial and final states, dictated by the binding energy of the inner-shell electron that gets expelled. Due to variations in binding energy among elements, the energy of characteristic X-rays differs for each element as well.

Characteristic X-rays are generated in various scenarios, such as within X-ray tubes, during nuclear reactions, and in particle accelerators. They are generated in the atmosphere when cosmic rays collide with atoms in the air. In X-ray imaging, characteristic X-rays arise when an X-ray photon collides with an atom, ejecting an inner-shell electron, after which the atom emits a new X-ray photon with a specific energy.

Characteristic X-rays possess greater intensity compared to bremsstrahlung X-rays; however, they are also more directional and constitute a minor fraction of the total X-ray emission. They are helpful in determining a material's composition since each has an energy unique to a particular element.

1.1.4 X-rays filtration

To improve image quality and decrease radiation exposure for patients, low-energy X-rays are filtered out of the X-ray beam through "X-ray filtration." The low-energy X-rays are removed by passing the beam through a filter that absorbs them while permitting the high-energy X-rays to pass.

For X-ray filtration, numerous filters can be used, such as:

- **Aluminum Filter**: This type represents the simplest form of X-ray radiography filters. They are placed in front of the X-ray detector and are therefore composed of a very thin layer of aluminum. High-energy X-rays penetrate aluminum filters, while low-energy X-rays, which are weaker and less efficient for imaging, are obstructed.
- **Copper Filter**: Despite being made of copper, these filters resemble aluminum ones. For greater filtration, copper filters are preferred over aluminum filters as they more effectively remove low-energy X-rays.
- **Multi-colored filters**: These filters are designed to selectively block specific X-ray energy levels. They are used when it is desired to merely exclude specific energy ranges from the X-ray beam, such as during the imaging of particular tissue types.
- **Grid Filters**: These filters are made up of several slender lead strips arranged at right angles to the X-ray beam. They are used in situations where radiation exposure must be reduced and to absorb additional low-energy X-rays beyond the previously mentioned filters.

The selection of the filter depends on the imaging method, the tissue being examined, and the factors related to radiation safety. X-ray filtering enhances image quality by removing low-energy X-rays, reducing scatter radiation and noise, and minimizing patient radiation exposure, thereby decreasing the likelihood of radiation-related damage.

1.2 Radiations:

1.2.1 What are Radiations?

Energy can be emitted or transmitted in various forms through space or a material medium. This creates different types of radiation

i. Electromagnetic Radiation:

This includes a wide range of waves, such as radio waves, microwaves, infrared waves, visible light, ultraviolet rays, X-rays, and gamma rays. These waves carry energy and travel through space at the speed of light

ii. Particle Radiation:

Particle radiation involves the emission of particles with energy. Examples include alpha radiation (helium nuclei), beta radiation (electrons or positrons), proton radiation (protons), and neutron radiation (neutrons). These particles can have different levels of penetration and ionizing abilities

iii. Acoustic Radiation:

Acoustic radiation are the propagation of energy in the form of sound waves. This includes ultrasound, used in medical imaging, and seismic radiation, associated with earthquakes and ground vibrations.

iv. Gravitational Radiation:

Gravitational radiation manifests as gravitational waves.

1.2.2 Types of Radiation

On the basis of energy of radiated particle, radiation is further divided in to two types:

- Ionizing Radiation
- Non-ionizing Radiation

A) Ionizing Radiation:

Ionizing radiation has the capability to strip electrons from atoms, resulting in the ionization of those atoms. It carries energy exceeding 10 electron volts (eV), a level sufficient to ionize atoms and molecules and disrupt chemical bonds. This type of radiation, often referred to as high-energy radiation, exerts various effects, but its most significant impact lies in the damage it can cause to DNA.

It is further categorized into two types:

1. Direct Ionizing Radiation:

It happens when electrons, which are charged particles with enough kinetic energy, interact with atoms or molecules, resulting in the formation of ions. This process is referred to as the direct process because the components interact directly without involving any intermediate steps. Examples include alpha particles or beta particles resulting from radioactive decay, as well as any other subatomic particles possessing sufficient kinetic energy to be ionizing.

2. Indirect Ionizing Radiation:

It occurs when uncharged particles interact with electrons in order to remove them from the atom. Example: X- rays and Gamma-rays are the common form of indirect ionization.

B) Non-Ionizing Radiation:

The energy of non-ionizing radiation is insufficient to directly extract electrons from their orbits. Nonetheless, electrons can be efficiently excited to higher energy levels by it. They typically release heat, which can occasionally be so strong that it can cause burns. Examples include ultraviolet, infrared, and visible light, among others.

1.3 Interaction of Radiation with Matter

The passage of radiation through towel results in either scattering or immersion, depending on the commerce between the radiation and matter. Specifically, electromagnetic radiation or photons parade three distinct modes of commerce with matter, primarily determined by their energy situations.

1.3.1 Photoelectric Effect

The energy of any electromagnetic radiation that strikes a metallic surface is transferred to the metal's electrons, which are likewise scattered. Once the material is lit, photo electrons are incontinently expelled. It's a wonder of low energy.

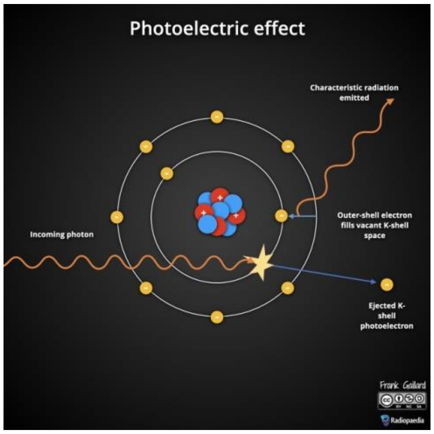

Figure 1 Illustration of photoelectric effect

1.3.2 Compton Effect

The Compton Effect consists of the interaction between a photon and a free electron situated in the outer orbital, resulting in the scattering of both the photon and electron in various directions. This method acts as the main vehicle for the exposure to ionizing radiation in radiotherapy. Notably, the absorption of incoming radiation is similar for both bone and soft tissues.

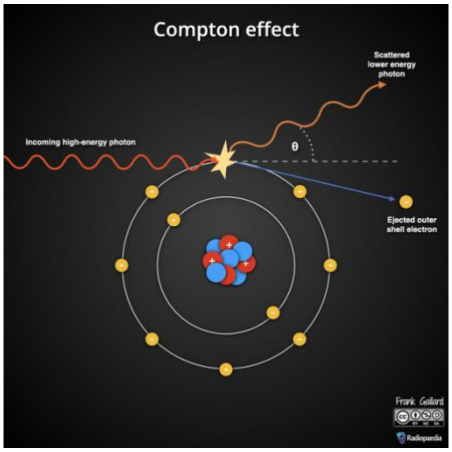

Figure 2 - Illustration of Compton's effect

1.3.3 Pair Production

The interaction of a high-energy photon with a heavy nexus may lead to the creation of an electron and a positron. The energy level of the photon for the brace product needs to be 1.02 MeV; below this threshold, the product will not manifest.

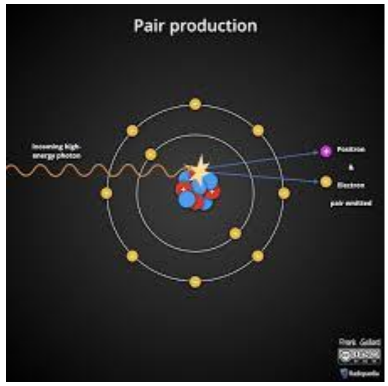

Figure 3 Illustration of pair production

1.4 Radioactivity

The discovery of radioactivity by Henri Becquerel in March 1896 is renowned as one of the most significant accidental findings in the field of physics. While studying phosphorescent minerals that exhibit light emission after exposure to sunlight, Becquerel observed that phosphorescent Uranium salts possessed the ability to spontaneously emit radiation that had the capacity to darken photographic plates. Radioactivity is the inherent characteristic of certain elements or isotopes to undergo the spontaneous emission of energetic particles or radiation, such as electrons or alpha particles, through the disintegration of their atomic nuclei. The radioactive disintegration reactions are of first order because in this rate of disintegration depends on the concentration term of radioactive material only. Gamma rays, alpha particles, and beta particles are some of the particles that are emitted during radioactivity.

- When an atomic nucleus undergoes decay, it emits two protons and two neutrons, referred to as alpha particles. They possess a positive charge and are relatively large and heavy. Alpha particles can be stopped by a sheet of paper or several centimeters of air because they do not deeply penetrate materials.

- During beta decay, beta particles, which can be electrons or positrons, are emitted. They are much smaller and lighter than alpha particles, and they carry either a negative or positive charge, depending on the situation. Beta particles can penetrate materials more thoroughly than alpha particles.

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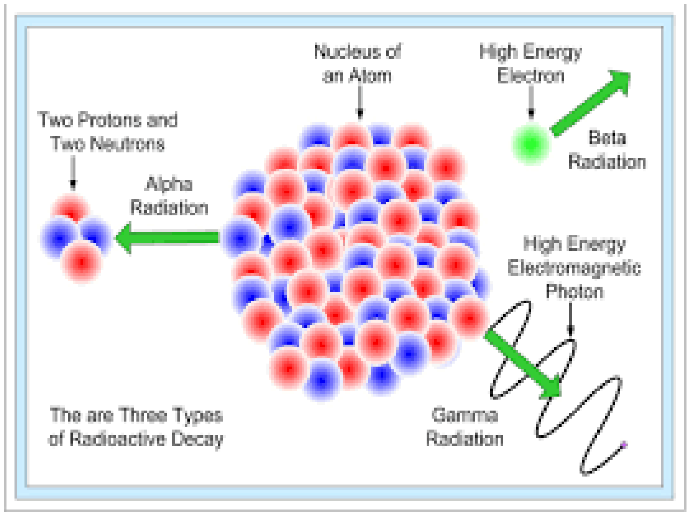


Figure 4 Radioactivity

- Gamma rays, which are a form of electromagnetic radiation, represent the more intense and penetrating type of radioactivity. They could penetrate matter much further than alpha or beta particles since they possess no charge and no mass. A meter of concrete or a few centimeters of lead do not obstruct gamma rays from passing through.

Comprehending radioactivity is crucial for various applications, such as nuclear power production, radiation treatment for cancer, and identifying radioactive substances for industrial and medical purposes. Contact with significant levels of ionizing radiation can be harmful to living beings, and safety measures must be implemented to guard against excessive radiation exposure.

1.4.1 Radioactive Isotopes

A radioactive isotope is a variant of an element that possesses an unstable nucleus and releases radiation while transforming into a more stable version. These isotopes may occur naturally or be produced artificially in a laboratory. They can serve multiple functions, including medical imaging, cancer treatment, and radiometric dating.

RADIOACTIVE ISOTOPE	APPLICATION IN MEDICINE
Cobalt-60	Radiation therapy to prevent cancer
Iodine-131	Locate brain tumors and monitor the activity of the heart, liver, and thyroid.
Carbon-14	Study metabolism changes for patients with diabetes, gout and anemia
Carbon-11	Glucose can be labeled and used to track organ activity during a PET scan.
Sodium-24	Study blood circulation
Thallium-201	Determine damage in heart tissue, detection of tumors
Technetium-99m	Radiotracers are employed in medical diagnostics to locate brain tumors and identify damaged heart cells.
Fluorine 10 (beta positive emitter)	Cancer diagnosis and staging

Table 1-Application of Radioactive isotope in medicine

The above mentioned radioactive isotopes are mainly use for treatments.

- **Radiation therapy:** High-energy rays target the tumor to destroy cancer cells.
- **Brachytherapy**: A minimal quantity of radioactive substance is positioned within or adjacent to the tumor. This technique is also referred to as "internal radiation therapy."
- **Cobalt-60:** It is likewise utilized in Radiotherapy to prevent cancer.
- **Fluorine-18:** It is utilized for diagnostic purposes in devices such as PET-CT and MRI.

1.5 Radiography:

1.5.1 What is Radiography?

Radiography involves using radiation to produce images of the organs, bones, and vessels within the body. Radiologists, specialized medical professionals skilled in interpreting these images, evaluate and diagnose the conditions shown. The success of patient care depends significantly on the precision and quality of the generated radiographic images.

1.5.2 The Role of a Radiographer

A radiographer plays a crucial role in generating numerous images that radiologists rely on for diagnosing patient conditions. It is essential that the specific body part of the patient is positioned correctly, and only the minimum necessary radiation is used to obtain a high-quality image.

Two key responsibilities of a radiographer should be highlighted: the proficient use of technology and the consideration of patient conditions within the healthcare environment. Radiographers conduct examinations and procedures on patients of all ages, including children and the elderly. They also work in various settings beyond the radiology department, such as in surgery, emergency rooms, cardiac care, intensive care, and patient wards. The expertise of a radiologic technologist is vital in aiding the diagnosis of injuries and illnesses.

1.6 Dental X-Ray

Dental X-rays (radiographs) are pictures of your teeth that dentists use to evaluate your dental health. These X-rays use minimal radiation to obtain clear images of your teeth's interior and adjacent structures.

Dental X-rays help dentists detect problems like cavities, tooth decay, and teeth cavity. Although they might appear intricate, they are actually standard and vital tools, equally significant as regular teeth cleanings.

1.6.1 Why are Dental X-Rays Performed?

Dental X-rays are typically conducted on an annual basis, although they may be performed more frequently if your dentist is monitoring the progression of a dental issue or treatment. Various factors can influence the frequency of dental X-rays you records.

1.6.2 Factors that influence the frequency of dental X-rays include:

- Age
- Current oral health status
- Any signs of oral disease
- A history of gum disease (gingivitis) or tooth decay

A new patient receives dental X-rays to provide dentist with a comprehensive understanding of his dental health. This is particularly crucial if he does not have any previous X-ray records from his former dentist.

Children may require dental X-rays more frequently than adults, as their dentists often need to monitor the development of their adult teeth. This monitoring is essential to determine whether baby teeth should be extracted to avoid complications, such as adult teeth emerging behind baby teeth.

1.6.3 Risks associated with dental X-rays:

Although dental X-rays involve exposure to radiation, the levels are minimal and are generally regarded as safe for both children and adults. If your dentist utilizes digital X-rays rather than traditional film, the risk of radiation exposure is further reduced.

To reduce unnecessary radiation exposure to essential organs, the dentist will cover your chest, abdomen, and pelvic region with a lead apron. In instances of thyroid disorders, a thyroid collar can also be utilized. Furthermore, children and women of reproductive age can wear both the lead apron and the thyroid collar.

Pregnancy stands out as a significant exception. Pregnant women or those who think they might be pregnant should avoid all forms of X-rays. Informing your dentist if you think you are pregnant is crucial, as radiation is considered unsafe for developing fetuses.

1.6.4 Preparation for Dental X-rays:

No special preparation is needed for dental X-rays. However, it is advisable to brush your teeth prior to your appointment to ensure a clean environment for the dental professionals working in your mouth. Typically, X-rays are conducted before dental cleanings.

During your visit to the dentist, you will be seated in a chair and provided with a lead vest that covers your chest and lap. The X-ray machine will be positioned near your head to capture images of your oral cavity. Some dental offices have designated rooms for X-rays, while others may perform them in the same area as cleanings and other treatments.

1.6.5 Types of X-rays:

There are various types of dental X-rays that provide different perspectives of your mouth. The most frequently used are intraoral X-rays, which include:

1. Bitewing:

This technique involves biting on a specific sheet of paper, enabling the dentist to evaluate the alignment of your tooth crowns. It is often used to identify gaps between teeth.

2. Occlusal:

This X-ray is captured with your mouth shut to assess the positioning of your upper and lower teeth. It can additionally recognize structural abnormalities in the palate or the mouth's floor.

3. Periapical:

This technique captures images of two complete teeth, from the root to the crown.

4. Panoramic:

In this kind of X-ray, the device revolves around your head

5. Extraoral:

These X-rays are employed when there is a suspicion of problems in areas beyond the gums and teeth, such as the jaw.

A dental hygienist will support you during the X-ray procedure. They might step out of the room for a short time while the pictures are being taken. You will be asked to stay motionless while the images are being captured. If film holders (spacers) are utilized, they will be positioned in your mouth to guarantee the accurate images are captured.

1.6.6 Following dental X-rays:

Once the images are available—immediately in the case of digital X-rays—your dentist will examine them for any irregularities.

If a dental hygienist is performing a teeth cleaning, the dentist may discuss the X-ray findings with you after the cleaning is completed, unless the hygienist identifies any significant issues during the X-ray process.

Your dentist detect any concerns, such as cavities or tooth decay, they will outline your treatment options. Conversely, if no issues are found, continue maintaining your excellent oral hygiene.

1.6.7 The perspective:

Regular dental X-rays, much like brushing and flossing, are essential for your overall oral health.

While a positive checkup can be reassuring, it does not eliminate the need for ongoing X-rays.

Depending on factors such as your age, health status, and insurance plan, X-rays may be recommended every one to two years. It is important to keep your appointments and consult your dentist promptly if you experience any discomfort or changes in your oral condition.

1.7 Capturing the Skull's Side View

Dental X-rays are commonly used to obtain a lateral view of the skull, offering valuable insights into the bone structure and alignment of teeth and jaws.

1.7.1 Radiation Safety in Dental Practice

While dental radiographs involve minimal radiation exposure, it is crucial to follow radiation safety protocols to ensure the well-being of both patients and dental staff.

1.7.2 Radiation Dose in Dental Imaging

Type of X-Rays	Estimated Radiation Dose(μSv)	Equivalent Background Radiation
Bitewing Radiograph	5 μSv	Comparable to 1 day of natural exposure
Periapical Radiograph	5–10 μSv	Equivalent to 1–2 days of natural radiation
Panoramic (OPG)	10–20 μSv	Equals 2–4 days of background radiation
CBCT Scan	20–200 μSv	Matches 4–40 days of natural exposure

Table 2 Radiation Dose in Dental imaging

1.7.3 Measures for Radiation Protection

1. **Patient Safety**

A. Lead Aprons and Thyroid Shields:

It is used to safeguard vulnerable organs from scatter radiation.

B. Digital Radiography:

It lowers radiation dose when compared to traditional film-based systems.

C. Limiting X-ray Usage:

It Performes only when clinically justified to prevent unnecessary exposure.

2. **Dental Team Precautions**

A. Maintain Distance (6 Feet or More): Helps reduce direct exposure.

B. Protective Barriers: Use of lead-lined shields for extra defense.

C. Radiation Monitoring Badges (Dosimeters): Track and assess long-term radiation exposure.

3. Considerations for Pregnant Patients

A. X-rays should be avoided during pregnancy unless essential.

B. If imaging is required, enhanced shielding with lead aprons and collars is mandatory.**[1][2][21]**

Chapter no Two

Literature Review

2.1 Study#1

2.1.1 Abstract:

This study compared the dose distribution and position verification distribution characteristics in diagnostic X-ray of Radiology Department and Dentistry department. A total of 35 cases of workers working in both departments were selected. The DIS dosimeter was read every two months and results were automatically saved on the instadoseTM platform. There is a gross deficiency of radiation monitoring in many of the facilities in the country. During the years of study, special dosimeters were used to quantify the radiation doses within the years 2013-2016. The outcome commemorated that levels of radiation received were safe and reduced over the years due to awareness and safety practices being put in place

2.1.2 Background:

X-rays have become a major component in the medical field, but pose a threat to the workers in cases where they are not well managed. In Nigeria, most hospitals do not have working systems of monitoring radiation dose and amongst those that do, very little is done in the analysis of the information. Various types of dosimeters exist which are intended for the capturing of radiation, and for this particular study, focus was directed to a modern known as DIS. The purpose is to quantify the radiation dose of hospital workers and assess the safety of the radiation dose.

2.1.3 Method:

The radiation exposure associated with the use of X-ray machines in dentistry and radiology was analyzed from 2013 to 2016. The study covered 35 radiology and dental staff members who operated X-ray machines. The HEAD DIS dosimeter was mounted on the subjects and their exposure was asses every two months. Statistical tools were used to analyze the gathered information. The readings were taken of deep, shallow and lens dose. Deep dose measures at a tissue depth of 1 centimeter while shallow dose measures of 0.007 centimeter like skin. Lens dose applies to 0.3 centimeter tissue of eye.

2.1.4 Conclusion:

The mean (total) dose in radiology department for the first, second, third and fourth year was 0.17 ± 0.08 (3.52) mSv, 0.08 ± 0.03 (0.77) mSv, 0.07 ± 0.04 (0.72) mSv and 0.07 ± 0.05 (0.55) mSv and in Dentistry was 0.08 ± 0.02 (0.73) mSv, 0.05 ± 0.02 (0.42) mSv, 0.05 ± 0.02 (0.24) mSv and 0.07 ± 0.04 (0.34) mSv; respectively

The radiation exposure for hospital personnel was below the international accepted benchmark. There was a reduction in radiation exposure levels over the four years due to increased safety culture. Workload did not correspond directly with radiation exposure. Overall mean dose in both department for occupationally exposed personnel were below IAEA annual dose limit of 20 mSv. This study emphasizes the need for regular radiation surveillance to protect health workers.

2.2 Study # 02

2.2.1 Abstract:

This research analyzes the current variety of dosimeters in use for dentistry. It assesses the combined outcomes of various studies performed with these dosimeters. The goal of radiographic technique optimization is to get the best quality imaging possible with the lowest radiation exposure to the patient. Ionization chamber dosimeter systems, film dosimeter, luminescence dosimeter, semiconductor dosimeter, alanine/electron paramagnetic resonance dosimeter, plastic scintillator dosimeter, diamond dosimeter, gel dosimeter, optically stimulated OSD, and thermo luminescent dosimeter (TLD) were all scrutinized for their accuracy and precision concerning the amount of radiation they measure. This study also mentioned the new developments in dosimeter I such as Instadose™ which provides instantaneous readings of radiation exposure. With modern dosimeters, particularly those implemented in cone beam computed tomography (CBCT), the radiation doses were found to be lower than previous levels while still giving accurate results. It was concluded that the relevant dosimeters and imaging techniques are selected appropriately to ensure the radiation safety in a dental practice.

2.2.2 Discussion:

Types of dosimeters:

The report summarizes each of the dosimeters used in dental radiography and their effectiveness in radiation measurement for patient safety. Dosimeters covered include:

- Ionization Chamber Dosimetry: This has high accuracy, but it needs electricity and several corrections to work effectively.
- Film Dosimetry : Requires a darkroom while offering 2D resolution.
- Thermo luminescent Dosimeters: Commonly made of calcium fluoride or lithium fluoride, these TL dosimeters are point dose readers which are small in size.
- Optically Stimulated Luminescence (OSL) Dosimeters: They provide immediate reading and release stored energy using light in place of heat.
- Semiconductor Dosimeters: They have very small size and are capable of providing instant readings, although calibration has to be done very precisely.
- Gel Dosimetry and Plastic Scintillator Dosimeters: These scintillator dosimeters are utilized in more advanced and research level radiographic examinations for high precision dose mapping.

This report also includes studies focused of estimation of exposures associated with dental intraoral radiography, panoramic radiography and CBCT. As an example, Mortazavi et al. noted an entrance surface dose (ESD) of 0.072 ± 0.019 mGy for 66 kVp OPG while Ludlow et al. reported effective doses for panoramic imaging as 16.1 μSv to 569 μSv for large field of view CBCT. In another study by Qu et al, it was found that use of thyroid collars resulted in a 48.7% reduction in thyroid gland dosage.

2.2.3 Result:

The results obtained show that new developments made in PLC clinical dosimetry, especially in CBCT, allows for much lower doses of radiation to be used without losing the quality of the diagnosis for the patient. The study strengthens the need of adhering to the ALARA (As Low As Reasonably Achievable) concept to improve radiation protection in dental radiography.

2.3 Study #03

2.3.1 Abstract:

Albania has had over a century of development on dental radiology. Medical imaging is an integral part of practicing dentistry and X-ray examinations assist a dentist in diagnosing conditions, planning them, and monitoring treatments or lesions. This research focused primarily on the profile of workers employed in dental radiation compared to those in diagnostic radiology and their respective occupational radiation exposure for the years 2016 to 2020 in Albania. The Institute of Nuclear Physics Applied conducted the assessment and analysis procedures and personal dosimetry laboratory. The population of monitored dental radiation workers rose from 39 in 2016 to 82 in 2020. Additionally, the annual collective effective exposure dose observed had fluctuated from 26.84man-Sv to 70.14man-Sv and appeared to have an upward trend. Furthermore, the average annual effective exposure dose for the total radiation workers varied between the range of 0.73mSv to 0.87mSv. The population of tooth diagnostic radiology workers increased from 196 in 2016 to 332 in 2020. The collective effectively annual exposure dose with a fluctuation between 240.99man-Sv to 358.03man-Sv also reveals an upward trend. The mean annual effective exposure dose from 1.08 mSv to 1.4mSv were during the period of monitoring postulated for diagnostic radiology. The acquired annual dose measurements were considerably lower than the internationally set dose level of 20 mSv.

2.3.2 Discussion:

The group involved in dental radiation comprised fewer workers than the diagnostic radiology group. In the dental radiation group, there were more women than men, and the women experienced a higher average annual increase than the men. The yearly combined exposure dose of dental radiation workers is less than that of diagnostic radiology workers.

For the average annual effective exposure dose of all monitored workers in the two groups of medical personnel assessed here, dental radiation workers usually recorded a lower average annual effective exposure dose compared to diagnostic radiology workers, who exhibited a higher average annual effective exposure dose.

Adhering to the concepts of radiation protection, exposure of all occupational exposure workers should be maintained at the level as low as practicably feasible (ALARA). Thus, radiographic procedures have to be optimized to deliver acceptable diagnostic information to the dentists at minimum, reducing radiation exposure to all Dental patients and workers.

2.3.3 Conclusion:

Annual collective effective dose of exposure vibrated from 26.84man-Sv to 70.14man-Sv, and the mean yearly effective dose of all monitored dental radiation workers oscillated from 0.73mSv to 0.87mSv. Annual collective effective exposure dose varied from 240.99man-Sv to 358.03man-Sv, and the mean annual effective exposure dose of total monitored diagnostic radiology workers ranged from 1.08 mSv to 1.40 mSv. The risk of radiation exposure of dental radiation workers is much smaller than that of diagnostic radiology workers.

2.4 Teeth Anatomy:

There are 32 teeth in the permanent dentition. This is composed of four incisors, two canines, four premolars or bicuspids, four molars and two wisdom teeth or third molars, in each of the jaws. If wisdom teeth have been extracted there will be 28 teeth.

2.4.1 Types of Teeth:

1. **Incisors:**

The four middle teeth on the top and bottom jaws are called Incisors. They are used to cut, tear and grasp food. The biting surface of an incisor is broad and thin that provides a chisel-shaped cutting edge.

2. **Canines:**

Positioned adjacent to the incisors, there are four canines (two above, two below).

They are sharp and assist in ripping food. They are referred to as "eye teeth" because of their location beneath the eyes.

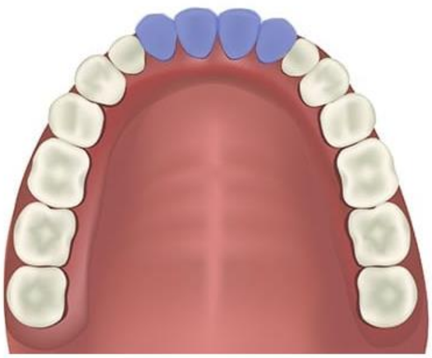


Figure 5 Incisors

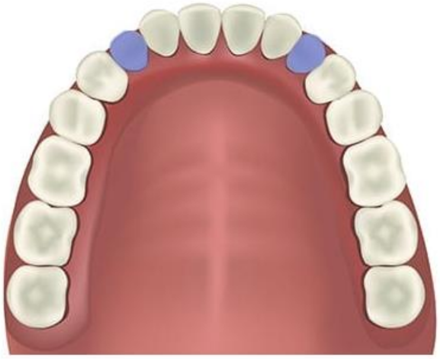


Figure 6 - Canines

3. **Premolars:**

They are situated behind the canines and in front of the molars, these teeth consist of two on each side of the jaw (four altogether). They possess a level surface featuring ridges, intended for crushing and grinding ingredients.

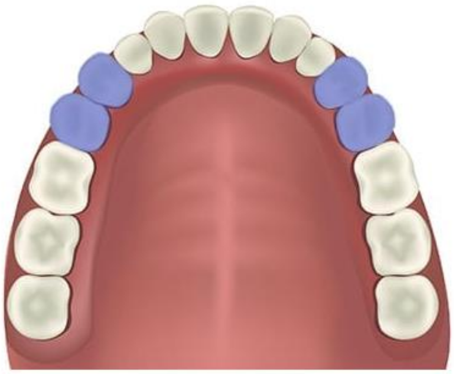


Figure 7 – Premolars

4. **Molars:**

These are the biggest teeth, found at the rear of the mouth. Usually, there are 12 molars, which encompass the wisdom teeth.

They feature a wide, flat area with cusps (sharp points) that assist in breaking down food into smaller fragments for ingestion.

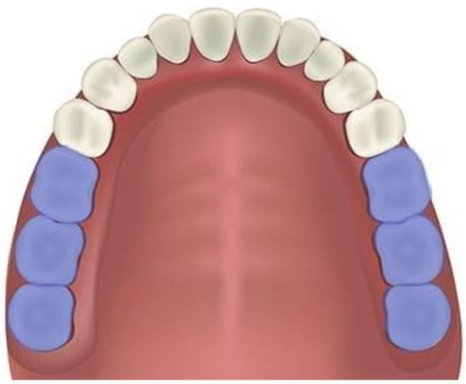


Figure 8 Molars

Wisdom Teeth:

Wisdom teeth are referred to as third molars. They break through from the age of 18 years and above but usually are removed surgically.

2.4.2 Structure Of Tooth:

The tooth consists of two anatomical structures, the crown and the root.

1. **Crown:**

The portion of the tooth that is visible above the gum line. It consist of the sub parts:

- **Enamel**: The hardest, outermost layer of the tooth that is highly mineralized. It is Primarily made up of calcium phosphate as hydroxyapatite. It Shields the tooth against erosion and acid damage.
- **Dentin:** It is present beneath the enamel. It is yellowish and softer than enamel. It is made up of tiny tubules that convey signals (such as pain or temperature variations) to the nerves within the tooth. It forms the majority of the tooth composition.

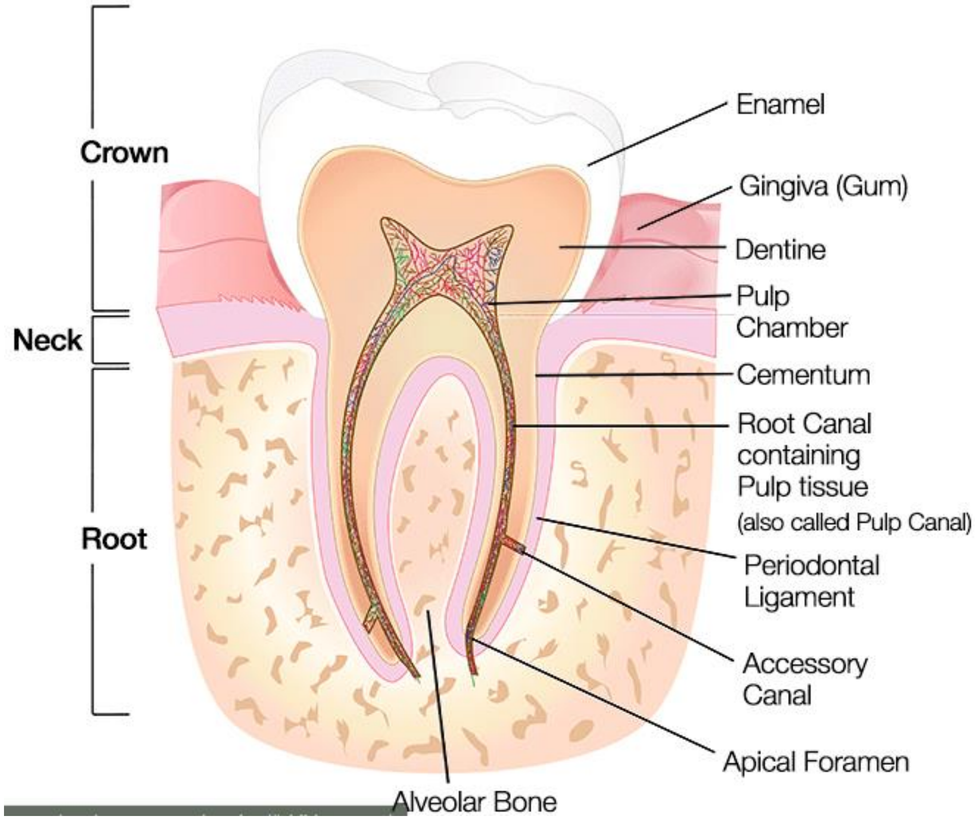


Figure 9 – Structure Of Tooth

- **Pulp Cavity :** The pulp chamber is the central most part of the tooth, lying under the dentine and running from the crown to the tip of the root. The deepest section of the crown. It includes dental pulp (nerves, blood vessels, and connective tissues). It is essential for dental health and sensory capabilities.
- **Gingiva:**
  The gingiva is the rose-colored soft tissue that we refer to as gums. It protects the jaw bone (alveolar) and tooth roots, and covers each tooth's neck.

2. Root:

A root of a tooth falls below the gum line, into the jawbone above or below, which supports the tooth within the mouth. Various categories of teeth also contain a varied amount of roots and root structure. Incisors, canines and premolars usually have a single root while molars would contain two or three.

- **Cementum**

Cementum is a hard tissue layer that covers the root of the tooth. It is nearly as hard as bone but much softer than enamel. It serves as the attachment surface for the **periodontal ligament**, which connects the tooth root to the **jawbone (alveolar bone)** and **gums**, keeping the tooth stable in its socket.

- **Root Canal (Pulp Canal)**

The root canal is the hollow space inside the root of the tooth that extends from the pulp chamber. It contains **blood vessels and nerves** that enter from surrounding tissues. It acts as a passage for these nerves and vessels to nourish the **pulp** and transmit sensory signals.

- **Periodontal Ligament**

This is a group of connective tissue fibers, known as fascicles, that surround the root of the tooth. One end of each fiber connects to the **cementum**, while the other attaches to the **jawbone.** It functions like a **shock absorber**, helping the tooth withstand pressure during biting and chewing, while also holding it securely in place.

- **Accessory Canal**

These are small branches that extend from the main root canal through the dentin to reach the **periodontal ligament**. It is most often found near the tip (apex) of the root. It provides additional routes for blood vessels and nerves to enter the pulp, supporting its nourishment and sensitivity.

- **Apical Foramen**

A small opening located at the very end (apex) of the tooth root. It allows **nerves and blood vessels** from the surrounding tissues to enter the root canal, maintaining the **vitality** and **sensation** of the tooth.

- **Alveolar Bone**

This is the part of the **jawbone** that surrounds and supports the roots of the teeth. It forms the **tooth socket** (alveolus) where each tooth is anchored. It works alongside the **periodontal ligament** to keep the tooth stable and secure in its position**. [18][22]**

Chapter no Three
X-cd machine, F-2011 Survey Dosimeter and X-ray room Analysis

BEST X-CD

3.1 Introduction

3.1.1 Why was intraoral equipment required?

In order to evaluate and assess the radiation distribution we used an intraoral device. With the help of a dosimeter we based our project on the radiation measurement, by altering the radiation levels through different settings. The primary goal was to reinforce guidelines within safety limits of dental procedure.

3.1.2 Why was BestX-CD used?

We opted for BestX-CD due to its consistency and accuracy in dental dose monitoring. Our project aimed at evaluating exposure levels and hence in was crucial to select an equipment which provided reliable functioning and precise radiation production. The space efficient design and easy to use structure made it essential for our needs, providing easy and prompt adjustments for diverse radiation settings.

Our project relied heavily on data acquisition, the BestX-CD assisted in the required radiation dose while we concentrated on assessing radiation doses through the dosimeter. We were able to observer and track exposure levels accurately due to the customizable exposure settings including, exposure period, electrical current, and voltage levels. It was possible to gather consistent data as the device had a uniform operation. Through this it was easier to test the dosimeters reliability and verify that the radiation dose stayed within safe thresholds for dental operations.

3.2. Device Description

3.2.1 Overview of the BestX-CD

Predominantly designed for diagnostic imaging, the BestX-CD, is a modern imaging equipment. Healthcare and dentistry facilities require the development and assembly of top-notch X-ray imaging which is provided by Best X Medical Technologies: a system which supports such a setup. The BestX-CD delivers everyday X-ray procedures, providing reliable image standard and improved safety elements, thus deigned to offer reliable and optimal output.

3.2.2 Key Components

1. **Monoblock (Tube Head)**

Figure 10 - Monoblock

- The most essential section in the BestX-CD diagnostic unit is the Monoblock or tube head. The X-rays are needed for radiography, which are produced in the X-ray tube enclosed in the unit. X-ray generation is a technique where electrical signals are changed into X-rays inside the tube head.
- The X-ray tube contains an anode which captures electrons when they strike together and a cathode which generates electrons. Energy waste is reduced and maximum charge flow is guaranteed by containing the entire unit in vacuum.
- The monoblock layout is space-saving. It minimizes excess exposure of radiation to zones outside the intended region due to integrated features such as collimators which direct the X-ray beam.

2. **Timer Unit**

- The duration of the exposure of the x-ray tube is governed by the timer unit. It calculates the duration of the radiation beam being produced in the diagnostic procedure.
- The quantity of the radiation dose administered to the patient has to be has to be managed within its time settings. Imaging dose for a particular x-ray process has to be has to be altered which is done by the technician since the timer is generally customizable. To ensure specific regulation of the exposure level the timer is set to milliseconds.

- The BestX-CD timer device may also incorporate protective mechanisms to ensure that the radiation time is within the limits, assisting to shield the patient from radiation overload.

Figure 11 - Timer Unit

3. Support Arm

- The support arm of the BestX-CD permits motion and placement of the imaging tube due to its physical framework that links the tube head to the main unit.
- The targeted adjustments provide ideal scan positions since the arm of the X-ray tube is built for movement and adaptability. The X-ray tube allows repositioning as required during operations, permitting up-down, side-to-side and turning movements.

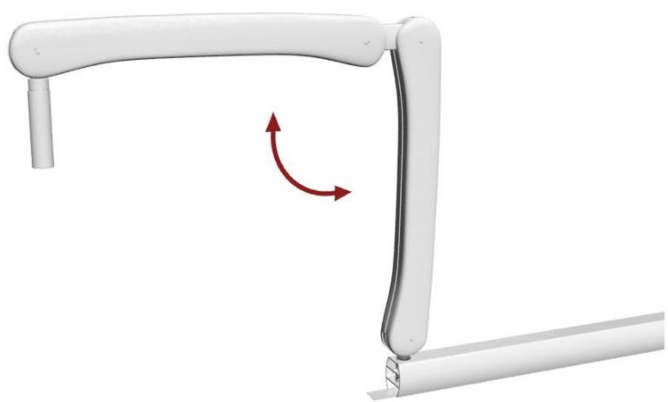


Figure 12 Support Arm

- The support arm is designed using robust substances to handle the load and the working pressure of the equipment, including secure systems to hold the imaging head steady once properly positioned.

4. Remote Control

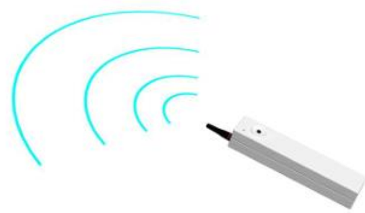


Figure 13 Remote Control

- The remote control adjusts different parameters required for imaging such as exposure time, voltage tube and current, from a safe distance adhering to the protocols.
- It is easy to handle due to being light weight and wireless, which further aids in maneuverability.
- An additional feature which minimizes excessive radiation is that the X-rays are emitted as long as the user is pressing it continually.

3.3 Function in the Project

3.3.1 How was it used in the FYDP?

In our FYDP (Final Year Design Project), the X-ray apparatus was used as the central source of radiologic emission to mimic real-life intraoral dental scanning environments. The primary goal was to replicate typical radiation contact and not to acquire clinical scans. This assisted in examining exposure pattern and scattering.

We managed the radiographic unit under standard parameters used in dental procedures including typical radiation doses, tube potential (Kvp) and electrical current (mA), to maintain actual radiation release. Exposures were made using standard procedures, as the equipment was arranged as it would be in a real-life intraoral radiographic process. Uniform and recurring radiologic emissions were generated, since the layout allowed us to create managed conditions.

X-rays were focused and excessive exposure was reduced, by using the equipment's collimation and X-ray positioning functions. This was done to mimic how oral health practitioners enhance imaging in healthcare setups. Furthermore, the layout enabled us to reposition the orientation and trajectory of the radiation beam for multiple trial placements, using the imaging head units' maneuverability.

In general, the X-ray apparatus operated as an essential component of our study, functioning as a dependable and steady X-ray emission generator. We were able to analyze radiation dose distribution, protection factors, and the impact of emission regulation methods since the operation allowed us to simulate actual oral scanning conditions.

3.3.2 Type of images produced (2D intraoral)

The X-ray machine produced two-dimensional scans. It images provided even minor details on the tooth structures and bone anatomy. This eventually lead to the diagnosis of tooth infections, cavities and bone abnormalities

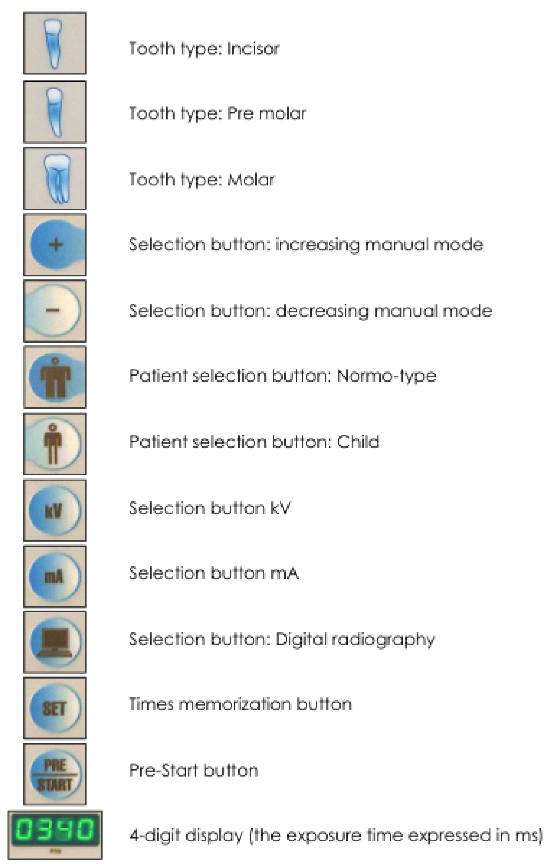


Figure 14 **General Symbols**

3.4 Key Specifications

3.4.1 Tube Voltage

Inside the X-ray tube, a potential difference is created between the anode and the cathode. This difference creates a voltage which is typically measure in Kvp. The penetrating ability and the energy is controlled by the voltage power, i.e. the higher the voltage the greater the penetration inside the tissue and bones. The imaging quality or the contrast in scans is also determined by the voltage. The adjustment in the voltage settings also affects the reader shown on the dosimeter. The BestX-CD provides voltage reliability up to ± 5 %.

3.4.2 Exposure Time Range

The time duration during which the X-ray radiations emit is known as the exposure time range. Typically, it is the duration the patient is exposed to the radiations. Exposure times is measured in seconds or milliseconds. The relation of the exposure time is directly proportional to the quality of the image being produced. Additionally, it also effects the dosimeter readings. The BestX-CD provides the reliability up to ± 5 %.

3.4.3 Focal Spot Size

Focal spot size is the area of the X-ray beams falling onto the anode through the cathode. The size and the shape of the focal spot is determined by the size and shape of electron beams respectively.

Head:	Monoblock with potential constant 100KHz model Toshiba DG-073B-DC; Toshiba D-045; Kailong KL11; Kailong KL21
Type:	60-70 kV$_p$ selectable (±5%)
Feeding voltage:	230 V~ (50 Hz) monophase
Maximun Power absorbed from the net:	0,800 KW
Cone diameter:	60 mm
Max symmetrical rays field:	Ø60 mm to SSD 200 mm
Anodic current:	4 mA – 7 mA selectable
Anodic voltage:	60, 70 kV selectable
Emission time:	from 20 mS to 3000 mS scale R10
Exposition time:	predefined
Max anodic voltage:	70 kV
Reference current time:	7mAs to 70 kV, 7mA, 1 s
Max supplied power:	0,49 KW a 70 KV, 7mA
Maximum apparent resistance of the power supply:	0.44 Ω
Weight:	6 kg

Accuracy Load Factors

Voltage accuracy	± 5 %
Current accuracy	± 3 %
Time accuracy	± 5 %
Dose accuracy	± 5 %

Figure 15 Key Specifications of Best X-CD [Ref: X-dc brouchure]

The shorter the focal spot the sharper the image it creates yet, the larger the focal spot the more heat it can withstand. The focal spot for BestX-CD is typically 0.4mm to 0.8mm.

FS-2011 Survey Dosimeter

3.5 Introduction to Dosimeters

Radiation safety is a top priority in dental clinics, not only for the patients but also for the dental professionals working with X-ray machines. Dental professionals, including dentists, hygienists, and radiographers, are frequently exposed to ionizing radiation during diagnostic procedures such as X-rays. To monitor and manage this exposure effectively, dosimeters are employed.

A dosimeter is a small, portable device that measures the amount of radiation a person is exposed to over a given period of time. It's an essential tool in radiation safety because it provides real-time data on exposure levels, helping ensure that dental workers are not exceeding safe radiation limits. For this study, the FS2011+ personal dosimeter was used.

The FS2011+ dosimeter is a compact, easy-to-use device that provides accurate, real-time measurements of radiation exposure in units of microSieverts per hour (μSv/h). This specific dosimeter is particularly well-suited for monitoring personal exposure to ionizing radiation, making it an ideal choice for dental professionals who require frequent and accurate monitoring during their work. The real-time readings on the device's display help professionals keep track of their exposure levels throughout the day, while also providing cumulative dose data, which is critical for long-term monitoring.

In addition to providing dose readings, the FS2011+ is a vital tool for ensuring compliance with safety standards. It helps dental clinics stay within the recommended exposure limits set by national and international safety guidelines, ensuring both staff and patient safety. Calibration, often performed by institutions like PINSTECH (Pakistan Institute of Nuclear Science and Technology), ensures the accuracy of these readings, maintaining reliability over time.

3.6 Functionality and Working Principle

The FS2011+ personal dosimeter is designed to measure and monitor radiation exposure, providing a clear and efficient way to track dose levels throughout a workday. Understanding how this device works is key to appreciating its role in ensuring radiation safety for dental professionals.

3.6.1 How the FS2011+ Works

The FS2011+ dosimeter operates by detecting ionizing radiation—the type of radiation emitted by X-ray machines during diagnostic procedures. Ionizing radiation has enough energy to remove electrons from atoms, which can damage or kill cells. The dosimeter detects this radiation by using an internal ionization chamber that senses changes caused by incoming radiation. When the radiation interacts with the chamber, it creates a small electric charge that the dosimeter measures.

The device then displays the dose rate in microSieverts per hour (μSv/h), which indicates how much radiation the person is exposed to over time. The FS2011+ can also accumulate total exposure, displaying this as microSieverts (μSv), which helps track cumulative exposure over longer periods.

3.6.2 Real-Time Monitoring and Data Logging

One of the main features of the FS2011+ is its ability to provide real-time measurements of radiation exposure. The dosimeter continuously updates its reading, allowing the wearer to stay informed about their current exposure. This capability is particularly useful in a dynamic environment like a dental clinic, where radiation levels may fluctuate during procedures.

The real-time display also enables dental professionals to take immediate action if they observe exposure levels trending higher than expected. For example, if a dental professional notices an unusually high reading during a specific procedure, they can adjust their positioning or ensure that the X-ray equipment is functioning correctly to minimize exposure.

Additionally, the device stores cumulative exposure data, which can be accessed later. This is valuable for both daily monitoring and long-term health tracking, as dental professionals are advised to keep records of their exposure for safety compliance purposes.

3.6.3 Calibration for Accuracy

The FS2011+ dosimeter is calibrated to ensure that the radiation measurements it provides are accurate and reliable. Calibration is typically performed by PINSTECH (Pakistan Institute of Nuclear Science and Technology) or similar accredited institutions. These organizations calibrate the device to international standards, ensuring that the dosimeter can measure doses accurately in terms of mSv (milli Sieverts) or μSv (microSieverts).

Regular calibration ensures that the device maintains its accuracy and is able to measure radiation exposure correctly, providing trustworthy data for radiation safety monitoring in dental environments.

3.7 Types of Dosimeters

In the realm of dental radiation safety, dosimeters come in various forms, each designed to meet specific needs. The FS2011+ dosimeter is one of the most commonly used personal dosimeters for monitoring radiation exposure in dental professionals. However, there are other types of dosimeters that may also be utilized in dental settings, depending on the specific application and needs of the clinic.

3.7.1 FS2011+ Dosimeter

The FS2011+ is a pocket-sized personal dosimeter that is ideal for real-time monitoring. This dosimeter is compact, lightweight, and easy to wear, making it highly convenient for dental professionals who need to monitor radiation exposure continuously during their workday.

Key Features of FS2011+:

- Real-Time Monitoring: Provides continuous updates of dose rate in µSv/h.
- Compact and Wearable: Small enough to be worn by dental professionals without interfering with their work.
- Data Logging: Stores cumulative dose data for long-term tracking of exposure.
- Easy Interface: Simple display with user-friendly buttons to navigate through different modes and settings.

This dosimeter is particularly useful in clinical environments where instant feedback is critical, as it allows workers to take immediate corrective action if radiation exposure levels become too high.

3.7.2 Thermoluminescent Dosimeters (TLDs)

- TLDs are another type of dosimeter that measure radiation exposure by using special crystals that absorb radiation. When the crystals are heated, they emit light in proportion to the amount of radiation they've absorbed. The amount of light emitted is then measured to determine the radiation dose.
- TLDs are often used in environments where long-term dose measurement is needed. Unlike the FS2011+, TLDs typically do not provide real-time data, but they are useful for periodic dose assessments over time.

3.7.3 Optically Stimulated Luminescence Dosimeters (OSLs)

- How They Work: OSL dosimeters work similarly to TLDs, but instead of heat, they use light to read the radiation absorbed by the crystals. The crystals are stimulated with light, and they release energy as light, which is then measured to determine the dose.
- Usage: OSL dosimeters are commonly used for long-term monitoring in environments where low-level radiation is present, as they offer high sensitivity and can be read multiple times.

3.8 Calibration and Accuracy

Accurate calibration is essential for ensuring that the readings provided by dosimeters like the FS2011+ are reliable and trustworthy. Calibration ensures that the device is correctly measuring radiation levels and adhering to international standards. Without proper calibration, the data provided by dosimeters could be inaccurate, leading to potential safety risks for dental professionals.

3.8.1 Calibration of FS2011+

The FS2011+ dosimeter is calibrated to internationally recognized standards to ensure the accuracy of its measurements. PINSTECH (Pakistan Institute of Nuclear Science and Technology), a well-known institution in radiation safety, typically conducts the calibration of dosimeters like the FS2011+. This calibration ensures that the dosimeter's readings are precise and meet the required regulatory standards, such as mSv (milliSieverts) or μSv (microSieverts).

3.8.2 The Calibration Process

The calibration process involves exposing the dosimeter to known levels of radiation in a controlled environment. This helps to establish a baseline or reference value for the dosimeter. Once calibrated, the device will accurately measure radiation doses within the specified range of exposure.

Regular calibration is critical to maintaining the accuracy and reliability of the FS2011+ over time. Calibration is typically done:

- Annually, to ensure long-term performance,
- Whenever the device is repaired or adjusted, to maintain precise measurements.

Calibration can also involve checking that the dosimeter is aligned with international safety standards, which may vary depending on the country or region.

3.8.3 Ensuring Accuracy and Reliability

The FS2011+ dosimeter is designed to be highly accurate, with an internal error-checking mechanism that helps ensure the reliability of its readings. Additionally, dental clinics using the dosimeter should ensure:

- Routine checks are carried out for battery life and sensor functionality.
- Proper storage is maintained, as exposure to extreme temperatures or humidity can affect the device's performance.
- Periodic calibration ensures that the device continues to give accurate and reliable readings, aligning with both national and international standards.

By regularly calibrating the FS2011+ and performing these routine checks, dental professionals can rely on the device for accurate radiation exposure monitoring, ensuring safety for both workers and patients.

3.9 Practical Use of FS2011+ in Dental Clinics

The FS2011+ dosimeter is a vital tool in real-time radiation monitoring within dental clinics. Its use ensures that dental professionals are continuously aware of their radiation exposure during diagnostic procedures, allowing them to take immediate action if exposure levels approach safety limits. This section will explore the practical applications of the FS2011+ dosimeter in a dental setting, focusing on how it is integrated into daily operations to enhance safety for both workers and patients.

3.9.1 Day-to-Day Use in Dental Procedures

In a typical dental clinic, the FS2011+ dosimeter is worn by the dental professional during all X-ray-related activities, including diagnostic imaging, radiography, and other radiation-producing procedures. Dental hygienists, assistants, and radiographers wear the device while taking X-rays, and sometimes even during procedures where radiation scatter may occur.

3.9.2 Key Benefits:

- Real-time monitoring: The dosimeter provides continuous dose rate readings, allowing dental professionals to be immediately aware of their exposure level throughout the procedure.
- Portability: The small, lightweight design makes it easy to wear throughout the day, without interference in routine tasks.
- Immediate feedback: The FS2011+ allows workers to immediately adjust their positioning, shield placement, or work practices if the exposure readings are higher than expected.

For example, if a dental hygienist notices an increase in dose rate during an X-ray session, they can adjust the patient's position or change shielding methods to reduce exposure. The ability to monitor cumulative exposure is also important for long-term safety, ensuring workers do not exceed recommended exposure limits over time.

3.10 Monitoring Radiation Exposure During High-Risk Procedures

Certain dental procedures, especially those involving multiple X-rays or longer exposure times, may increase the risk of radiation exposure. The FS2011+ dosimeter provides ongoing dose assessments, even during more complex procedures like full-mouth radiographs or panoramic imaging, ensuring that the radiation dose remains within safe limits.

By continuously measuring the dose rate and cumulative exposure, the dosimeter offers a real-time safety check, alerting the operator when the radiation dose exceeds predefined safety thresholds. This allows dental clinics to stay compliant with health and safety regulations while minimizing radiation risks to workers.

3.10.1 Training and Awareness for Dental Staff

An important aspect of using the FS2011+ dosimeter effectively is ensuring that all dental staff are trained on how to use it. Dental professionals must be taught:

- How to wear the dosimeter properly during X-ray procedures,
- How to read the device's output, understanding dose rates and cumulative exposure readings,
- When to adjust work practices based on the device's readings (e.g., repositioning the patient, adjusting shielding, or modifying exposure settings on the X-ray machine).

By training staff to regularly check the dose rate and understand how to act upon it, the clinic can create a culture of radiation safety where both the dental professionals and patients are protected from unnecessary exposure.

3.10.2 Real-Time Monitoring for Staff and Patient Safety

The FS2011+ dosimeter helps to establish a safe working environment by continuously monitoring radiation exposure levels. It offers dental professionals a tool to actively manage radiation safety throughout the day. By providing real-time feedback, the dosimeter helps:

- Minimize the cumulative radiation dose to dental professionals, ensuring they remain within acceptable limits.
- Reduce patient exposure by adjusting procedures as needed, based on dosimeter readings.
- Increase accountability, with data logging for future review and audit purposes, ensuring compliance with safety regulations.

By integrating the FS2011+ dosimeter into everyday practice, dental clinics can optimize radiation safety protocols, minimize exposure risks, and ensure that radiation safety standards are continuously met. **[5]**

3.11 Safety and Regulatory Considerations

Radiation safety is a critical concern in dental radiography. The FS2011+ dosimeter plays a central role in ensuring the health and safety of dental professionals by continuously monitoring their radiation exposure. However, its use must also align with established safety protocols and regulatory standards to ensure that all staff are protected, and the clinic remains compliant with health and safety regulations.

3.11.1 Radiation Safety Protocols

To ensure the safety of dental professionals and patients, radiation safety protocols must be followed rigorously in any dental setting where X-rays are used. These protocols include:

- Limiting radiation exposure to the minimum necessary for diagnostic purposes (also known as ALARA, which stands for As Low As Reasonably Achievable).
- Using shielding: Both the patient and the dental professional should use appropriate shielding materials (like lead aprons) to minimize radiation exposure.
- Proper training: All dental staff should be adequately trained in radiation safety procedures, including how to properly use dosimeters like the FS2011+ to monitor their exposure levels.

The FS2011+ dosimeter plays a vital role in the ALARA principle by providing dental professionals with real-time data about their exposure levels. If exposure exceeds safe limits, the dosimeter alerts the wearer, allowing for immediate corrective action (such as repositioning or adjusting the X-ray settings).

3.12 Regulatory Standards

There are various regulatory bodies and standards that govern radiation safety in dental clinics. These include national organizations like the National Council on Radiation Protection and Measurements (NCRP) in the United States, and international bodies like the International Commission on Radiological Protection (ICRP). These agencies set exposure limits for workers who handle radiation.

Occupational Dose Limits:

- Regulatory bodies set specific limits for the amount of radiation that can be safely absorbed by workers in a year. For example, in the United States, the NCRP recommends an annual dose limit of 50 mSv for occupational exposure for radiation workers.

Patient Dose:

- Guidelines also exist for patient exposure during dental radiographic procedures. The goal is to minimize patient dose while ensuring that diagnostic quality images are obtained.
- The FS2011+ dosimeter is calibrated to meet these regulatory standards. Its accurate, real-time monitoring ensures that dental professionals do not exceed permissible radiation exposure levels, while also providing a means for clinics to remain compliant with safety regulations.

3.13 Regular Monitoring and Documentation

To comply with radiation safety standards, regular monitoring of exposure levels is essential. The FS2011+ dosimeter assists dental clinics in meeting this requirement by:

- Storing cumulative exposure data for review and auditing purposes.
- Tracking real-time dose readings to ensure that exposure does not exceed acceptable levels during procedures.

These capabilities provide clinics with documented proof that they are adhering to safety standards, which is crucial during inspections or audits by regulatory bodies.

3.13.1 Record Keeping and Compliance

Maintaining records of radiation exposure is a key component of regulatory compliance. The FS2011+ dosimeter helps dental clinics by automatically logging radiation exposure data over time. This data can be:

- Used to monitor staff safety and ensure no one exceeds the set exposure limits.
- Reviewed periodically during health and safety audits to ensure that proper radiation safety protocols are being followed.

By consistently recording exposure levels, the FS2011+ helps clinics stay compliant with the regulatory requirements that govern radiation use in medical and dental settings.

3.14 Case Studies or Practical Examples

To better understand the practical use of the FS2011+ dosimeter in real-world dental settings, let's examine a couple of case studies that highlight how this device enhances radiation safety for dental professionals. These examples demonstrate the importance of continuous monitoring and immediate feedback, which are made possible by the FS2011+.

3.14.1 Case Study 1: Real-Time Monitoring During a Full-Mouth X-Ray Procedure

In this case, a dental hygienist was preparing to take a full-mouth radiograph (FMX) of a patient, which typically involves taking 18-20 X-ray images. This type of procedure involves higher radiation exposure because of the number of images taken and the positioning of the patient.

- Dosimeter Use: The dental hygienist wore the FS2011+ dosimeter on their uniform while positioning the patient and operating the X-ray machine.
- Real-Time Monitoring: Throughout the procedure, the dosimeter provided real-time readings of the dose rate. When the hygienist noticed an increase in exposure during one of the more challenging angles, the FS2011+ indicated a rise in radiation exposure above

safe levels. This triggered an immediate adjustment of the X-ray settings and the patient's positioning.

- Outcome: By having the FS2011+ dosimeter, the hygienist was able to adjust their approach, reducing unnecessary exposure to both the staff and the patient. The dosimeter's continuous feedback allowed for proactive safety measures that helped ensure the procedure was completed safely.

Figure 16 FS-2011 Survey Dosimeter

3.14.2 Case Study 2: Monitoring Cumulative Radiation Exposure in a Radiography Clinic

In a busy radiography clinic with multiple operators, it is crucial to monitor the cumulative radiation exposure of dental staff throughout the day. This case involves a radiographer who wears the FS2011+ dosimeter during their shifts.

- Dosimeter Use: The radiographer wore the FS2011+ for a 6-hour shift, during which multiple X-rays were taken for a variety of dental procedures, including bitewings, panoramic images, and periapical radiographs.
- Cumulative Exposure Tracking: The dosimeter logged the total dose of radiation the radiographer was exposed to over the course of the shift. At the end of the day, the total cumulative exposure was reviewed to ensure that it stayed within acceptable limits.
- Outcome: The FS2011+ dosimeter provided accurate data that confirmed the radiographer's exposure was well within safety guidelines. The clinic could confidently report that radiation safety protocols were adhered to, and the radiographer's exposure was properly monitored.

This case highlights the importance of cumulative exposure tracking, especially in environments where multiple staff members are regularly exposed to radiation.

3.14.3 Impact on Worker Safety

Both case studies demonstrate how the FS2011+ dosimeter contributes significantly to worker safety by providing real-time feedback and accurate cumulative data. In both scenarios, the ability to monitor radiation exposure directly helped limit unnecessary exposure and ensured that dental professionals remained within safe radiation limits.

3.15 X-ray Room Analysis:

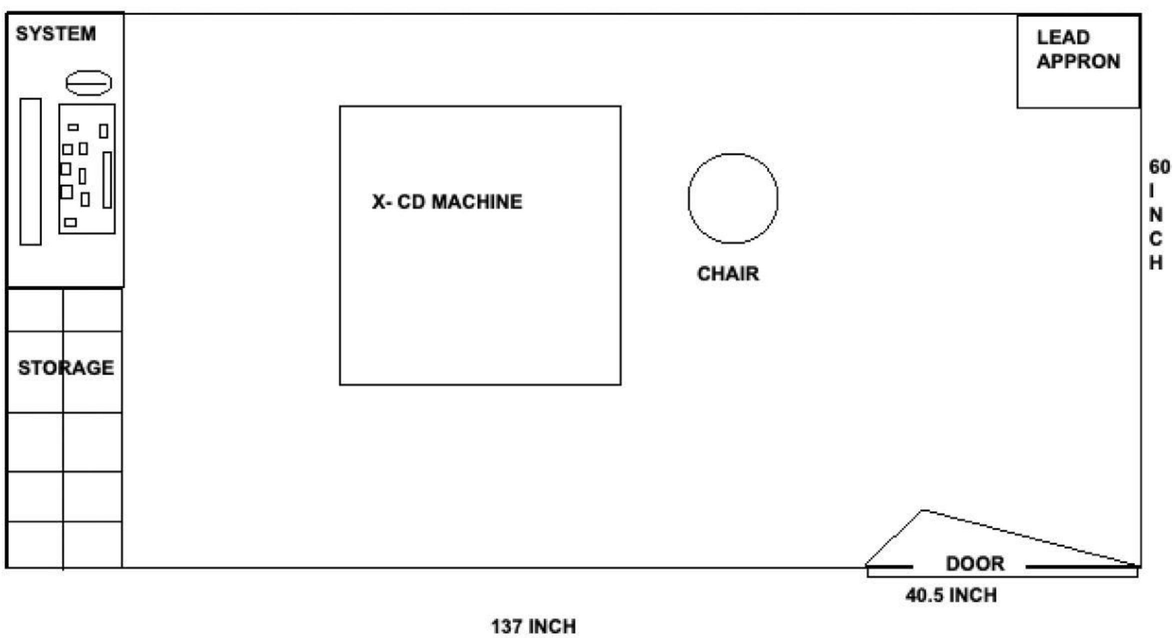


Figure 17 Design Layout of X-ray Room

3.15.1 Key components and their placement

- **Room Dimensions:**
  The room is 137 inches in length and 60 inches in width, providing adequate space for safe operation of X-ray equipment and movement of workers.

- **X-CD Machine:**
  The X-ray machine is centrally located in the room, ensuring the radiation source is well-positioned for procedures and measurements.

- **Chair:**
  Positioned close to the X-ray machine to hold the patient during procedures, ensuring alignment with the X-ray beam.

- **System Panel:**
  Located on the left wall for operating the X-ray equipment. It allows the operator to control the machine from a safe distance.

- **Storage:**
  Placed adjacent to the system panel for storing necessary accessories, such as X-ray films, protective equipment, or tools.

- **Lead Apron:**
  Located on the right wall for easy access. Lead aprons are critical for protecting workers and patients from unnecessary radiation exposure during procedures.

- **Door :**
  Positioned at an angle (40.5 inches in width), ensuring efficient entry and exit while maintaining shielding integrity.

Chapter no Four
METHODOLOGY

4.1 Objective:

The primary objective of this experiment is to measure the dose rate at varying kilo voltage (kVp), milli ampere (mA), angles, and distances. Additionally, we apply regression analysis to identify the relationship between these variables and the dose rate. By conducting this experiment, we aim to understand how different parameters affect the dose rate and use the regression model to predict future dose rates for different settings.

4.1.1 Dosimetric Quantities:

Dosimetric quantities refer to the physical parameters utilized to measure the radiation dose that a patient receives during radiotherapy. These quantities play a crucial role in ensuring that the patient is administered an optimal radiation dose that effectively treats cancer while reducing the likelihood of radiation-induced side effects. The primary dosimetric quantities employed in radiotherapy include: Absorbed Dose: This represents the energy absorbed by a patient's tissue per unit mass, measured in Gray (Gy).

- **Equivalent Dose**: The equivalent dose assesses the effect of radiation on biological tissue, taking into account factors such as the type of radiation, its energy, and the sensitivity of the tissue to radiation. It is expressed in Sievert (Sv) and serves as an indicator of the potential biological consequences of radiation exposure on a patient.
- **Effective Dose:** The effective dose provides a holistic measure of the radiation dose received by the entire body, considering the differing sensitivities of various organs and tissues to radiation. It is determined by multiplying the equivalent dose for each specific organ by a weighting factor that indicates the relative risk of radiation-induced cancer in that organ. The effective dose is also expressed in Sievert (Sv) and offers a standardized approach for evaluating the overall biological effects of radiation exposure on the human body.

These Dosimetric quantities are used to ensure that the radiation dose received by the patient is accurate, effective, and safe.

4.1.2 Terminologies:

1. **Radiograph:**

A radiograph is the image that has been created on a media such as film or sensitive plate due to the effects of x-rays, gamma rays, or any neighbouring form of radiation**.**

2. Intraoral Radiograph

Intraoral X-ray is an x-ray taken with the film located intra orally. It consists of bitewing, periapical, and occlusal view.

3. Extra oral Radiograph:

Extra oral x-ray is taken with the film positioned outside the mouth. It includes panoramic and cephalometric images.

4. Bitewing Radiograph:

A form of intraoral x-rays that encaptures the upper and lower sixty degrees bits of the teeth, x-raying cavity/bite in between usage most dominantly**.**

5. Periapical Radiograph:

It displays the tooth in full detail with its surrounding bone. It is useful in discovering problems of the root and the bone.

6. Occlusal Radiograph:

The radiograph displays the full arch of the teeth in either the upper or lower jaw.

7. Panoramic Radiograph (OPG):

Two-dimensional dental X-ray which displays the whole mouth in a single shot.

8. Cephalometric Radiograph:

A lateral view of the head taken for orthodontic purpose to check the alignment of the jaws and teeth.

9. Cone Beam Computed Tomography (CBCT)

CBCT is particularly useful in planning implants and performing root canal surgery as it captures the desired spatial regions of the bone in high detail.

10. X Ray Tube head:

This is where the x-rays are created within an x-ray machine.

11. Collimator:

Return of the x-ray cross section is a mechanism that reduces the x-ray scatter and patient exposure.

12. Filter:

Heats the disc using rays of x-direction in order for the low energy x-rays to be erased.

13. PID (Position Indicating Device):

X-ray emission cone is belonging to the X-ray apparatus unit and serving to direct the beam is called position indicating device.

14. Kilovoltage (kVp)

Regulates the charge of the beams penetrate to the tube.

15. Milliamperage (mA)

Regulates the amount of x-rays produced in a machine.

16. Exposure Time:

Interval during which x-rays are produced determines the density of the image.

17. Density:

The degree of blackening on a processed radiograph is referred to as density.

18. Lead apron:

Protective clothing that is used to cover the patient's body from unnecessary radiation.

19. Thyroid collar:

A protective lead shield placed at neck level to protect the thyroid gland from radiation.

20. ALARA principle:

"As Low as Reasonably Achievable" – reducing the overall radiation exposure to staff and patients.

21. Paralleling technique:

A technique where the film or sensor is placed parallel to the long axis of the tooth while the x-ray tube is perpendicular.

22. Bisecting angle technique:

A method in which the x-ray beam is aimed perpendicularly to an imaginary line bisecting the tooth and the film.

4.1.3 Occupancy Factor:

The occupancy factor denotes the duration during which a specific area or room is utilized by personnel or patients who might be exposed to radiation. This factor is crucial for ensuring radiation safety and protection. It is employed to determine the effective or equivalent dose that individuals receive while present in an area where radiation exists. The calculation considers both the length of time spent in the area and the radiation levels present.

Typically, the occupancy factor is represented as a fraction or percentage of the total time the area is occupied. It takes into account various elements such as staff working hours, patient treatment schedules, and the frequency of occupancy in a particular room or area.

For instance, in a dental department, the occupancy factor for x-ray rooms may vary based on the duration patients spend during procedures. If a treatment room is occupied by a patient for a considerable part of the day, the occupancy factor for that room would be higher than that of a room used for shorter treatment sessions. By integrating the occupancy factor into radiation safety calculations and dose evaluations, radiation therapists, medical physicists, and radiation safety officers can ensure the implementation of suitable radiation shielding measures and safety protocols. This approach aids in reducing radiation exposure for both patients and staff, thereby fostering a safe working environment.

The occupancy factors for various areas within the dental department can differ based on specific circumstances and usage patterns. Below are some examples of occupancy factors for common areas:

1. Control Room:

The control room serves as the operational hub for X-ray monitoring equipment, which is both operated and monitored from this location. Typically, the occupancy factor in the control room is high, as staff members dedicate a considerable amount of their working hours to this area. During operational hours, the occupancy factor for the control room can approach 75%.

2. Corridors:

Corridors are generally regarded as areas with low occupancy, where people move through instead of remaining for extended durations. The occupancy factor in corridors is often lower than in other spaces. This factor is influenced by the frequency and length of personnel or patient movement. The occupancy factor for corridors can vary from 10% to 20% or even lower, depending on the particular facility and its usage trends.

3. Outside Area:

The occupancy rate for the external area of a dental department, including the adjacent grounds or open spaces, is generally quite low. These locations are typically not utilized for long durations by either staff or patients. Consequently, the occupancy rate for the external area may be considered negligible or nearly 0%.

4. Parking Area:

The parking area serves as the designated location for staff, patients, and visitors to park their vehicles. Given that individuals typically spend a limited amount of time in this area, the occupancy rate is comparatively low. This rate is influenced by the dimensions of the parking area, the number of users, and the length of their stay.

5. Waiting Areas:

Waiting areas, including reception areas and patient waiting rooms, can exhibit different occupancy factors. These factors are influenced by elements such as patient volume, appointment scheduling, and the time patients spend in these spaces. The occupancy factor for waiting areas may vary from 10% to 25% or even higher, contingent upon the particular facility and patient flow. It is crucial to understand that these occupancy factors serve as general examples and may differ according to the specific practices and operational characteristics of each radiation therapy facility. A thorough assessment and evaluation of the actual occupancy factors should be performed by radiation safety officers and professionals to guarantee that suitable radiation protection measures and safety protocols are established.

4.2 Equipment used in procedure:

In this experiment, the following pieces of equipment and materials are needed:

- X-ray machine: This machine can be adjusted for both kvp and mA settings. These settings are proportional to the radiation produced during the exposure.
- Survey dosimeter: This device determines the radiation exposure measuring its ionization in a gas caused by X-rays.
- Phantom or Calibration Equipment: A phantom simulates human tissues in order to check how accurately the radiation measurements reflect the dose rate in a biological system.

- Protractor/Angle Measurement Tool: This instrument helps in placing the survey dosimeter at particular angles with respect to the X-ray setting e.g. (0° 45° 90° etc).
- Measuring tape or ruler: Used to set distances from the x-ray source for example 8inch, 16inch and 24inch to the survey dosimeter.
- X-ray control console: Panel that makes changes and sets the kVp and mA for the different trials for the experiment.
- Data Logging Equipment: Instruments or soft wares used to record and save the exposure values for analysis later.

4.3 Data Description:

The dataset for this study was generated through carefully controlled radiation dose measurements taken in a dental radiographic setting. The primary aim was to assess how the dose rate varies based on changes in machine settings and spatial factors such as distance and angular orientation from the X-ray source. A structured and systematic approach was used to ensure the readings captured a wide range of real-world clinical scenarios.

Measurements were taken under two standard exposure configurations:

- **70 kilovoltage peak (kVp) at 7 milliamperes (mA)** – representing a higher energy and intensity setting.
- **60 kVp at 4 mA** – representing a lower energy setup, which is also commonly used in routine dental radiographs.

To capture how the radiation spreads directionally, measurements were taken at **nine different angles** around the X-ray head. The angles included **0°, 45°, 90°, 135°, 180°, 215°, 270°, 315°, and 360°**. These angular positions provided a 360-degree overview of the radiation field, helping identify zones of higher and lower exposure. Each measurement involved careful positioning of the survey dosimeter using a protractor or angle guide to ensure precision.

For each setting, readings were taken at **three distances** (8, 16, and 24 inches) from the X-ray source to simulate varying proximity, and at **angles ranging from 0° to 360°** in 45° steps to assess directional variation in dose rate.

These settings were chosen because they reflect typical operating conditions found in dental clinics, especially during intraoral imaging procedures.

4.4 Data Collection:

4.4.1 Steps in the procedure:

1. First Preparations:

- Setting Up Equipment: Check that the X-ray machine, survey dosimeter, and all other components are set, calibrated, and in working order as per the manufacturer's guidelines. This also includes checking that the survey dosimeter is calibrated as per manufacturer specifications.
- Radiation Safety Procedure Verification: All radiation safety measures need to be checked. This includes the use of protective clothing such as lead aprons to reduce exposure as much as possible.
- Initial Settings for X-ray Machine: First, set the X-ray machine to the first exposure condition of 70 kVp at 7 mA.

2. Measurements of Angles and Distances: Positioning the Chamber:

- The first position of the survey dosimeter is at 0° and 8 inches away from the X-ray source.
- Take a dose rate measurement at this position. Then move the dosimeter to the next angle of 45° and repeat the measurement.
- This process is to be continued for all specified angles which include 90°, 135°, 180°, etc. At the distance of 8 inches.
- After completing the measurements at 8 inches, move the chamber to 16 inches and repeat the process.
- Finally, place the chamber at 24 inches and repeat the measurements for all angles.
- For 60 kVp at 4 mA, repeat: Once the measurements for the first set of exposure parameters (70 kVp, 7 mA) are completed, adjust the X-ray machine to the second set of exposure parameters (60 kVp, 4 mA) and perform the same procedure for all angles and distances.

All measurements were recorded in units of **microSieverts per second (μSv/s)**, which reflect the rate of radiation dose being delivered at each point. This unit is commonly used in occupational and medical radiation safety assessments, making the results directly applicable to clinical practi

4.4.2 Data Recording:

Record all the measured values of the dose rate in tabulated form, with the values being associated with the respective angles, distances, and exposure parameters.

Provide both the measured dose rates and the calculated dose rates from the regression model, and the error of each measurement.

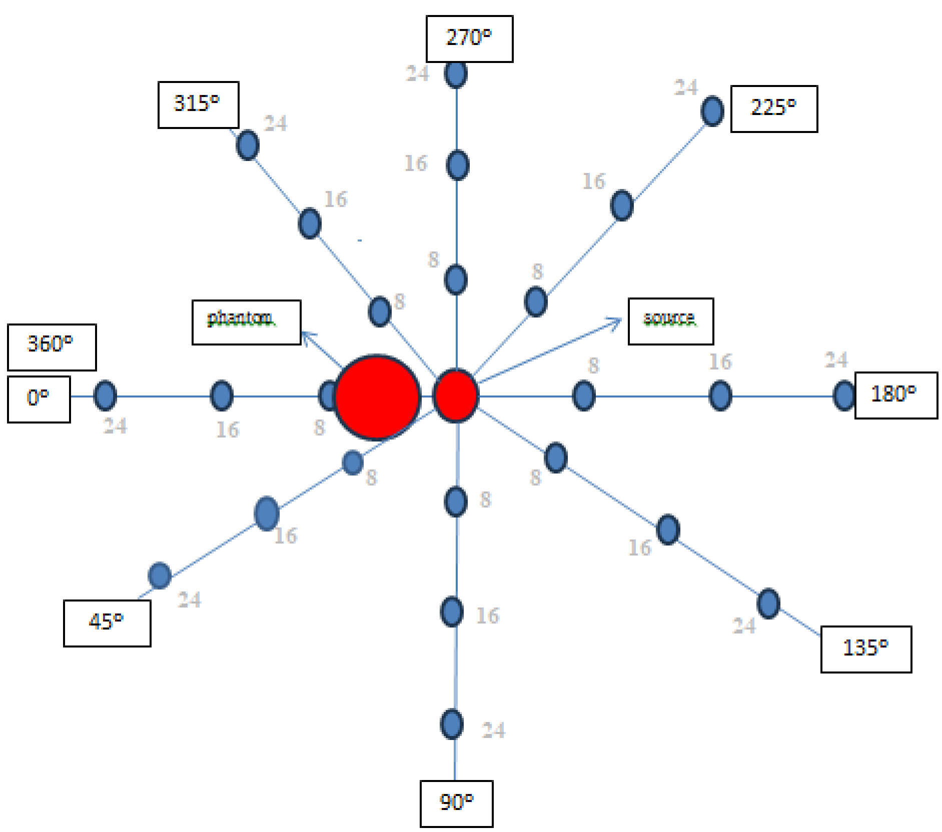


Figure 18 Angle Diagram

4.5 DATA ANALYSIS:

For data analysis, we are using statistical tool of linear regression.

4.5.1 Linear Regression:

Linear regression is a supervised machine learning algorithm that derives insights from labeled datasets and constructs linear functions that best fit the data points, which can then be employed for making predictions on new datasets.

It assumes that a linear relationship exists between the input and output, meaning that the output changes at a consistent rate as the input changes. This relationship is represented by a straight line.

4.5.2 Why Linear Regression is Important?

- **Clarity and Comprehensibility**:
  It is straightforward to grasp and interpret, thus serving as the basis for exploring machine learning.
- **Forecasting Capability:**
  It aids in predicting future outcomes based on historical data, making it applicable in various fields such as finance, healthcare, and marketing.
- **Basis for Additional Models:**
  Many advanced algorithms, including logistic regression and neural networks, are built upon the concepts of linear regression.
- **Effectiveness:**
  It is computationally efficient and performs admirably on problems exhibiting a linear relationship.
- **Broadly Utilized:**
  It is among the most frequently employed techniques in both machine learning and statistics for regression analysis.
- **Examination:**
  It provides insights into the relationships between variables (for instance, the extent to which one variable influences another).

4.5.3 Best Fit Line in Linear Regression

In linear regression, the best-fit line is the straight line that best represents the relationship between the independent variable (input) and the dependent variable (output). It is the line which reduces the discrepancy between the actual points and the model-predicted points.

1. Goal of the Best-Fit Line

The aim of linear regression is to identify a straight line that will reduce the error (difference) between the predicted values and the actual data points to a minimum. This line assists us in forecasting the dependent variable for new, unseen data.

Here Y is referred to as a dependent or target variable and X is referred to as an independent variable also called the predictor of Y. Regression can be accomplished using numerous forms of functions or modules.

2. Equation of the Best-Fit Line

For simple linear regression (with one independent variable), the best-fit line is given by the equation

$$y = mx + b$$

Where:

y = forecasted value (dependent variable)

x = the input (independent variable)

m =slope of the line (how much y changes when x changes)

b = y-intercept (the value of y when x = 0)

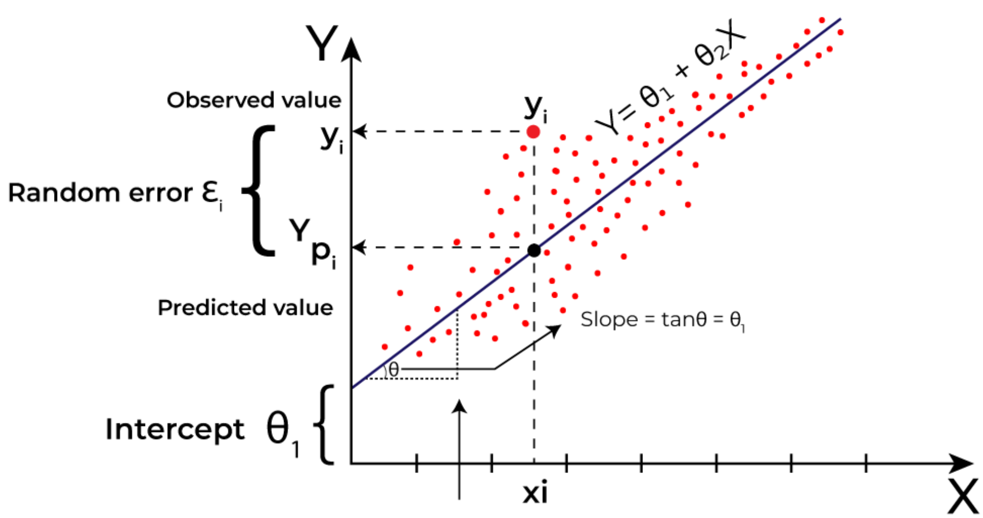


Figure 19 – Linear Regression Graph

The line that best fits the data will be the one that maximizes the values of m (slope) and b (intercept) that makes the predicted y-values as close as possible to the observed data points.

3. Minimizing the Error: The Least Squares Method

To determine the best-fit line, we apply a technique known as Least Squares. The principle of this method is to reduce the total of squared discrepancies between the actual values (data points) and the estimated values from the line to the least amount possible. These discrepancies are termed residuals.

Formula:

$$\text{Residual}=y_i-y_i'$$

Where $\mathbf{y_i}$ is the **observed value.**

$\mathbf{y_i}'$ is the **predicted value.**

The least squared method minimize the sum of the squared residuals:

$$\text{Sumofsquarederrors(SSE)}=\Sigma(y_i-\hat{y}_i)^2$$

4. Interpretation of the Best-Fit Line

This method guarantees that the line best reflects the data, minimizing the total of the squared differences between the predicted and actual values as much as possible.

Slope (m): The slope of the line of best fit indicates the change in the dependent variable (y) corresponding to a one-unit change in the independent variable (x). For example, if the slope is 5, then for every 1-unit increase in x, the value of y will rise by 5 units.

Intercept (b): The intercept represents the y value at which the line intersects when x equals 0. It is the point where the line crosses the y-axis.

In linear regression, certain hypotheses are assumed to ensure the reliability of the model's results.

4.5.4 Limitations

- The technique presumes the linearity of the relation between the variables. Linear regression may fail when the relationship between the variables is not linear.
- Outliers can significantly affect the slope and intercept, skewing the best-fit line.

In total, a large volume of readings was recorded, covering every combination of:

- Exposure setting (2 levels: 70kVp/7mA and 60kVp/4mA)
- Distance (3 levels: 8", 16", 24")
- Angle (levels: 90,135,180,225,270)

This resulted in **60 unique measurement combinations per exposure setting**, giving a comprehensive dataset for analysis.

4.6 Linear Regression Analysis

After visually analyzing the dose rate data, the next step was to apply a **linear regression model** to quantify the relationship between the dose rate (dependent variable) and the various independent factors influencing it. Linear regression is a foundational statistical technique used to model and evaluate the strength and direction of relationships between variables. In this case, the model was designed to determine how factors such as **kilovoltage (kVp)**, **milliamperage (mA)**, **distance**, and **angle** contribute to variations in dose rate.

4.6.1 Model Formulation:

The linear regression model was constructed using the following equation:

$$\text{Dose Rate} = \beta 0 + \beta 1\ (\text{kVp}) + \beta 2\ (\text{mA}) + \beta 3\ (\text{Distance}) + \beta 4\ (\text{Angle}) + \varepsilon$$

Where:

- $\beta 0$ is the intercept (baseline dose rate when all predictors are zero.
- $\beta 1$ *(beta-1)* to *$\beta 4$ (beta-4*) are the coefficients (slopes) that quantify the effect of each variable.
- ε is the error term capturing residual variation.

All input variables were treated as **continuous numerical values**. The dataset contained 50 valid observations after cleaning and organizing, with all relevant combinations of distance, angle, kVp, and mA.

Ways or terms used to give the goodness of fit of the model

1. Residuals of each predicted vs actual. Yactual – Ypred. Ideal redusla magnitude very low (near to zero) and should be random.
2. Value of R^2 . Ideally 100 %

4.6.2 Model Output and Summary Statistics

The linear regression was performed using statistical software, and the following key metrics were obtained from the model:

Metric	Value
R-squared (R^2)	0.737
Adjusted R-squared	0.713
F-statistic	30.20
P-value (overall model)	< 0.00000000001
Root Mean Square Error	11.76 μSv/s
Number of observations (n)	60

Table 3 Model output of linear regression

The R^2 value of 0.737 indicates that the model now explains approximately 73.7% of the variability in the dose rate values. This suggests a stronger fit, with the included predictors capturing more of the underlying pattern in the data.

The F-statistic and corresponding p-value confirm that the model is statistically significant overall ($p < 0.00000000001$), indicating that the predictor variables collectively have a strong influence on the dose rate.

4.6.3 Coefficient Interpretation:

Below are the regression coefficients and their interpretations:

Variable	Coefficient (β)	P-value	Interpretation
Intercept	+21.07	0.393	Not statistically significant; suggests the baseline dose rate has high variability.
Distance	–2.665	< 0.000001	Highly significant. For every unit increase (e.g., in distance), dose rate decreases by ~2.67 μSv/s.

Variable	Coefficient (β)	P-value	Interpretation
kVp	**+0.765**	**0.030**	**Statistically significant. A unit increase in this variable increases dose rate by ~0.77 μSv/s.**
mA	**+3.31**	**0.021**	**Statistically significant. A unit increase leads to a rise in dose rate by ~3.31 μSv/s.**
Angle	**–0.039**	**0.148**	**Not statistically significant; indicates a weak negative trend.**

Table 4 Coefficient Intrepretion

4.6.4 Key Observations:

- **Distance**: Distance showed a **strong negative correlation** with dose rate. The coefficient of –2.665 and the **highly significant p-value ($p < 0.000001$)** confirm that increasing the distance substantially reduces the dose rate, which aligns with the inverse square law of radiation.
- **kVp** : A **positive and statistically significant** relationship was observed (**$p = 0.030$**). The coefficient (+0.765) indicates that increasing the tube voltage increases the dose rate, which is consistent with the physics of X-ray production, where higher kVp results in more energetic photons and increased radiation output.
- **mA** : The results show that mA has a **statistically significant** effect on dose rate (**$p = 0.021$**), with a **positive coefficient (+3.31)**. This suggests that within the measured range, higher current increases the radiation dose rate, likely due to an increased number of emitted photons.
- **Angle**: Although the coefficient (–0.039) was negative, the **p-value (0.148)** indicates that a positive linear relationship, indicating that dose rate increased with increasing angle,

4.7 Residual Analysis and Model Limitations

- Residuals: Scattered residuals suggest some unexplained variance, potentially due to non-linear effects or interaction terms not captured in this linear model.
- Durbin-Watson Statistic: Not provided, but should be revisited to check for autocorrelation, especially if time-sequenced data was used.
- Normality Tests: The Omnibus and Jarque-Bera test results are not reported here, but assuming no major issues (as in the previous model), we can accept the assumption of normal residuals.
- RMSE: The reduced RMSE of 11.76 μSv/s indicates improved prediction accuracy, but still leaves room for refinement particularly at extreme parameter values.

- The linear regression model explains a substantial portion of the variation in dose rate ($R^2 = 0.737$) and identifies Distance, kVp, and mA as statistically significant predictors.

However, the presence of unexplained residuals and one insignificant variable indicates that future work could explore non-linear modeling (e.g., quadratic or interaction models) to better capture dose behavior in dental X-ray systems, especially at parameter extremes.

4.8 Evaluation of Quadratic Regression and Its Limitations

In our effort to build a comprehensive model that accurately represents how various physical and technical parameters influence radiation dose rate in dental radiography, we explored both linear and quadratic regression techniques. The decision to test a quadratic regression model was based on initial observations from graphical analysis, where patterns such as angular symmetry and the rapid decay of dose with increasing distance suggested potential non-linear relationships. Quadratic regression is commonly used in such contexts because it introduces squared terms, allowing the model to capture curvature and interactions that a linear model cannot.

To construct the quadratic model, additional variables were included in the dataset: squared terms for kilovoltage (kVp^2), milliamperage (mA^2), distance^2, and angle^2, alongside their linear counterparts. The expectation was that this model would provide a more flexible and precise fit, especially given the theoretically curved nature of radiation dispersion and the exponential behavior of energy output from X-ray tubes. All variables were formatted numerically, and regression was conducted using a structured Excel model.

However, once the quadratic regression model was executed, the results showed clear signs of model failure. Most notably, the **R^2 value was negative**, which is a statistically insignificant. A negative R^2 indicates that the model fits the data **worse than a simple horizontal line based on the mean** of the observed values. In practical terms, this means that using the quadratic equation to predict dose rate would result in greater error than simply assuming the average dose rate for all measurements making the model not just ineffective but actively misleading.

4.8.1 Model Output and Summary Statistics:

Metric	Value
R-squared (R^2)	–0.174
Adjusted R-squared	–0.408
F-statistic	Not significant
P-value (overall model)	> 0.05

Root Mean Square Error	High (not meaningful)
Number of observations (n)	60

Table 5 Model output of quadratic regression

There are several reasons why the quadratic regression failed in this context. First, the **number of independent variables increased substantially** due to the addition of squared terms, but the **sample size remained limited (n = 60)**. This imbalance likely led to overfitting, where the model attempts to conform to random noise in the data rather than capturing generalizable patterns. Over fitting reduces predictive accuracy and inflates error when the model is applied to new or unseen data.

Second, the inclusion of squared terms introduced **multicollinearity**, especially between variables like angle and angle^2 or distance and distance^2. When predictor variables are highly correlated with one another, it becomes difficult for the regression algorithm to assign meaningful weights, often resulting in unstable or nonsensical coefficients. This can distort the entire model and further reduce its reliability.

4.8.2 Coefficient Interpretation:

Below are the regression coefficients and their interpretations:

Variable	Coefficient (β)	P-value	Interpretation
Intercept	Unstable	> 0.05	Baseline dose not significant.
kVp	Fluctuating	> 0.05	No consistent influence detected.
mA	Inconsistent	> 0.05	Likely affected by narrow mA range and overfitting.
Distance	Unreliable	> 0.05	Multicollinearity with Distance^2.
Angle	Unstable	> 0.05	Not interpretable due to overlap with Angle^2.
kVp^2	Not significant	> 0.05	No additional predictive value.
mA^2	Not significant	> 0.05	Adds noise rather than clarity.

Distance²	Not significant	> 0.05	Insufficient to model dose falloff.
Angle²	Not significant	> 0.05	Theoretically valid but not supported by data.

Table 6 coefficient intrepretation

Third, it is possible that **measurement limitations or data variability** particularly at low-dose levels or far distances (e.g., 24 inches) added noise to the model. At these points, the dose rates were close to background levels, making them more sensitive to even minor changes in environmental conditions, phantom placement, or survey dosimeter alignment. These subtle inconsistencies may have impacted the model's ability to detect clean quadratic patterns.

4.9 Model Comparison and Final Model Justification

After conducting both graphical and statistical analyses, and testing multiple regression approaches, it became essential to compare the modeling techniques to determine which best represents the behavior of dose rate in dental radiography. Although both **linear** and **quadratic regression models** were initially considered, the outcomes of each differed significantly in terms of statistical validity, interpretability, and practical relevance.

The **linear regression model** included four independent variables—kilovoltage (kVp), tube current (mA), distance, and angle and provided a relatively strong fit to the measured data. It achieved an **R^2 value of 0.735**, meaning approximately 73.5% of the variation in dose rate was explained by the model. The model's Root Mean Square Error (RMSE) was **11.76 µSv/s**, and key variables such as distance, angle, and kVp showed statistically significant contributions to dose rate prediction. These results were not only statistically robust but also aligned with known physical principles in radiographic science particularly the inverse relationship between distance and radiation intensity, and the positive relationship between kVp and dose output.

In contrast, the **quadratic regression model** which aimed to capture non-linear behavior through the addition of squared terms for each predictor did not perform successfully. Despite its theoretical potential to model curved relationships (e.g., angular symmetry, exponential decay), the model returned a **negative R^2 value**, indicating that it failed to capture the variance in the data better than a horizontal mean line. This was a critical indicator that the model was not reliable. The likely reasons for this failure include a limited dataset size relative to the number of variables, multi collinearity between original and squared terms, and potential overfitting. Furthermore, the predicted values generated by the quadratic model deviated substantially from the actual dose rates, further confirming its poor predictive capability.

From a modeling perspective, it is not sufficient for a method to be theoretically appropriate; it must also be supported by the dataset it is applied to. In this study, the quadratic model introduced unnecessary complexity and statistical noise, while the linear model remained stable, interpretable, and predictive. The simplicity of the linear model proved advantageous: it allowed for direct understanding of how each input variable influenced the dose rate and provided reasonable estimates that could be used for clinical planning and radiation safety assessments.

Thus, after a full comparison, the **linear regression model was retained as the final and most appropriate model** for this study. It provided a statistically sound balance between accuracy, interpretability, and practical applicability, while the quadratic model despite being tested was rejected due to clear evidence of poor fit and unreliability.

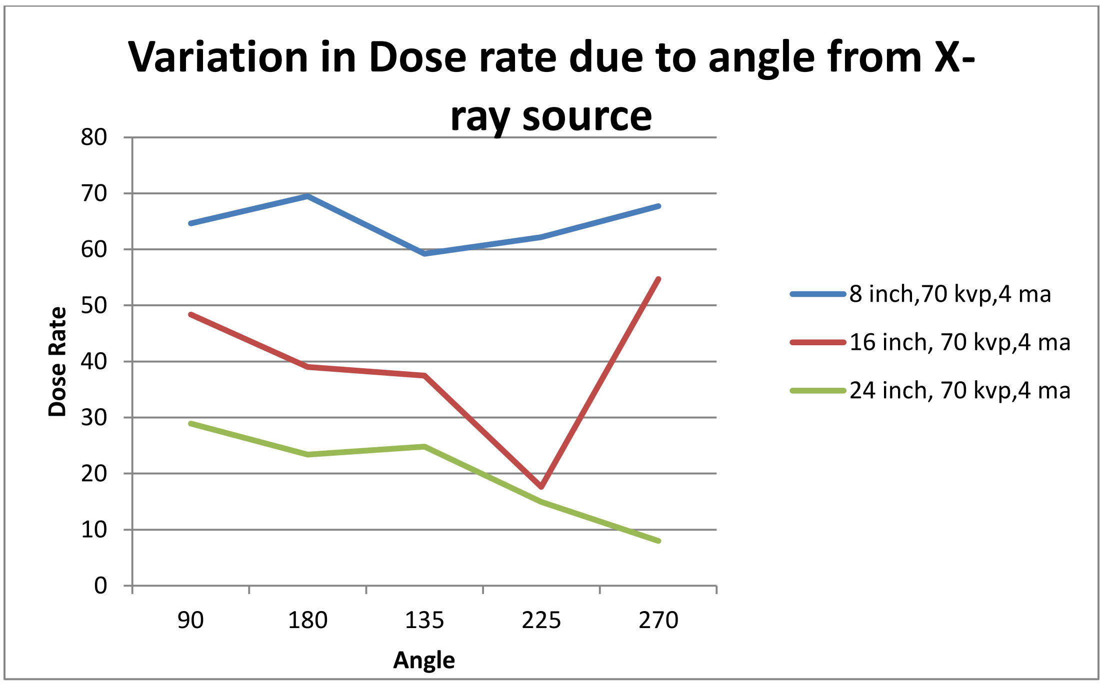
In conclusion, the analysis supports the use of a **linear regression approach** for modeling dose rate under varying machine settings and spatial parameters in dental radiographic environments.

Chapter no Five

Result and Discussion

5.1 Graphical Analysis:

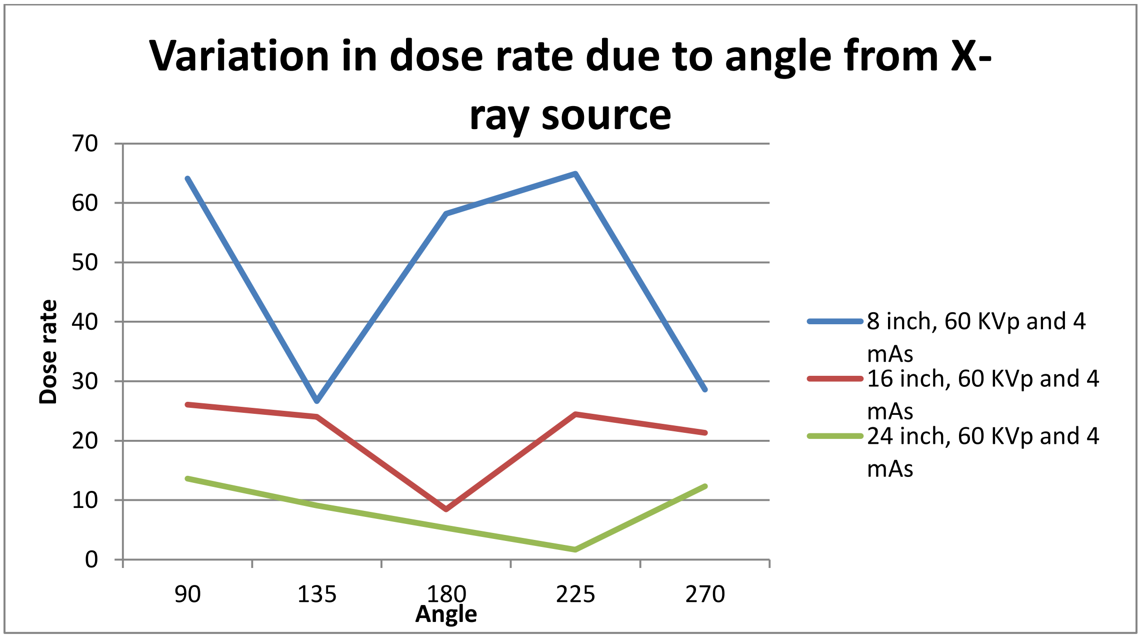
5.1.1 Angular Variation in Dose Rate (70 kVp, 4 mA)



1. This graph presents the variation in radiation dose rate with angle, measured at three fixed distances 8", 16", and 24" using a constant X-ray exposure setting of 70 kVp and 4 mA.
2. At all distances, it is evident that angular position significantly affects the distribution of radiation, as the dose rates do not remain consistent while the angle changes around the X-ray head.
3. At a distance of 8 inches, the dose rate remains largely stable; however, there are noticeable reductions at 180° and 225°, indicating variation based on direction. Even at close proximity, the observed angular trend suggests that the X-ray beam is primarily concentrated in a forward-facing direction, and there is measurable attenuation or shielding at certain angles.
4. At 16 inches, the angular variation is more distinct:

5. There is a steady reduction in dose rate from 90° to 225°, followed by a sharp increase observed at 270°.
6. This pattern suggests that certain angular positions receive a greater amount of scattered or indirect radiation, while others such as 225° may lie outside the main beam path or be partially shielded. The increased reading at 270° may reflect a more direct or reflected path of radiation, possibly influenced by surrounding environmental or positional factors.
7. At a distance of 24 inches, the angular trend becomes more gradual and less variable. Although overall dose levels are reduced, a consistent decline beyond 180° can still be observed, reinforcing the directional nature of the radiation field even at greater distances, the angle continues to influence how much stray or scattered radiation reaches the point of measurement.

5.1.2 Angular Variation in Dose Rate (60 kVp, 4 mA)

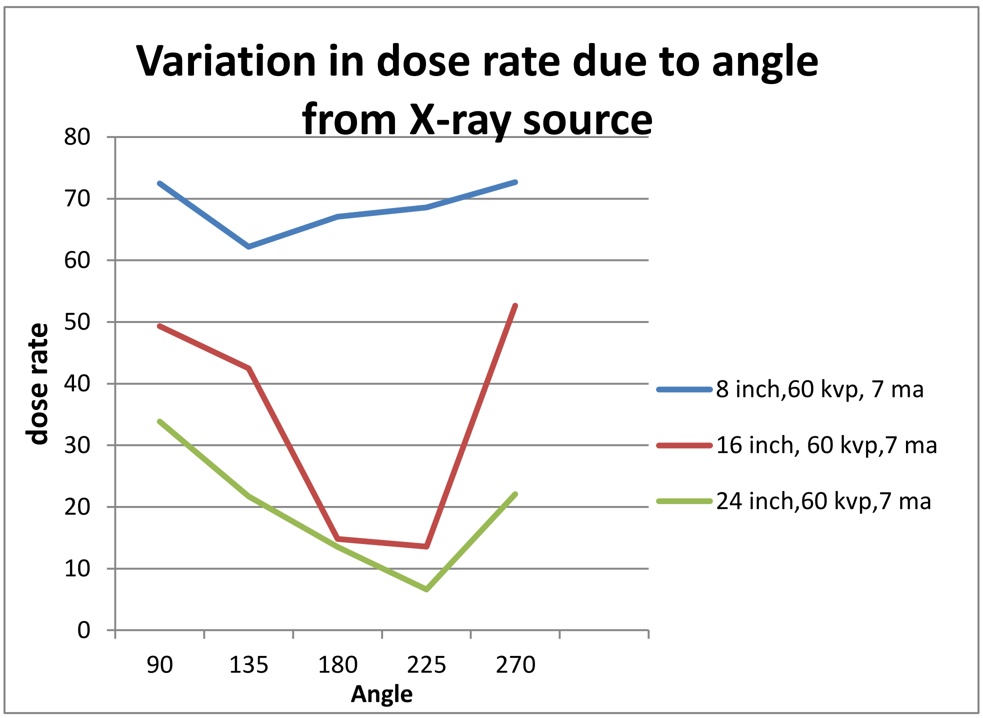


1. This graph presents the variation in radiation dose rate with angle, recorded at three distances 8", 16", and 24" using a constant X-ray exposure setting of 60 kVp and 4 mA.
2. At all three distances, it is evident that angle significantly influences dose distribution, as the dose rate fluctuates with changes in angular measurement around the X-ray beam path.
3. At a distance of 8 inches, the dose rate exhibits substantial variation. The highest readings are observed at 90° and 225°, while a noticeable reduction appears at 135° and again at

270°. This pattern suggests that the radiation beam is highly directional at close range, and variations in surrounding surfaces, shielding, or geometry may affect readings at certain angles.

4. At 16 inches, the angular differences are less pronounced but still observable:
5. There is a gradual decrease in dose rate from 90° to 180°, followed by a moderate rise toward 225° and stability through 270°.
6. This trend indicates that some angular attenuation is present, but the effect is more balanced than at shorter distances. The lower values at 180° suggest the presence of partial shielding or less direct exposure at the rear of the beam, while the elevation at 225° may result from reflected or scattered radiation.
7. At 24 inches, the dose rate values are overall lower, and the angular pattern is smoother. The data show a mild dip around 180°, with a slight increase toward 270°, indicating reduced but still present angular influence. Despite being farther from the source, the detector still captures directional differences in scattered or residual radiation.

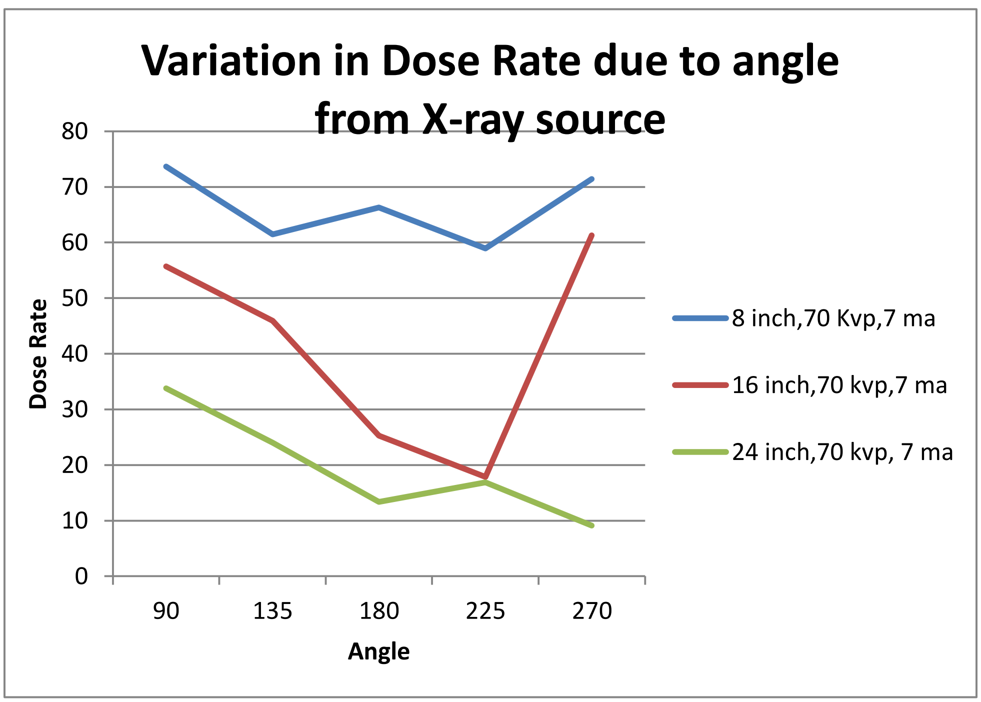
5.1.3 Angular Variation in Dose Rate (60 kVp, 7 mA)



1. This graph presents the angular variation in radiation dose rate measured at three distances 8", 16", and 24" using a constant X-ray exposure setting of 60 kVp and 7 mA.
2. The data indicate that both **angle** and **distance** influence the dose distribution pattern. The dose rate varies across angles at each distance, reflecting directional exposure characteristics typical of diagnostic X-ray equipment.

3. At 8 inches, the dose rate remains consistently high, with only minor fluctuations across the angles measured. A slight reduction is observed around 135° to 180°, followed by a gradual increase toward 270°. These results suggest that close proximity to the X-ray source results in a strong and stable exposure pattern, with minimal attenuation across angles.
4. At 16 inches, the angular trend is more pronounced:
5. The dose rate decreases steadily from 90° through 225°, reaching a minimum at 225°, and then rises significantly at 270°.
6. This behavior suggests directional beam fall-off combined with shielding or angular misalignment at 225°. The increased value at 270° may correspond to a more favorable alignment with the beam's scattering direction or reflective surfaces.
7. At 24 inches, the overall dose rate is reduced due to increased distance, but the angular pattern is similar to that observed at 16 inches. A progressive decline is evident from 90° to 225°, followed by an increase toward 270°. The reduction at 180° and 225° suggests lower exposure on the side opposite to the primary beam path, while the rise at 270° indicates enhanced exposure from scatter or secondary alignment.

5.1.4 Angular Variation in Dose Rate (70 kVp, 7 mA)



1. This graph displays the variation in radiation dose rate across different angular positions, measured at three distances 8", 16", and 24" under constant exposure settings of 70 kVp and 7 mA.

2. At all distances, a clear angular dependency is observed. The dose rate does not remain uniform across angles, reflecting the directional emission characteristics of the X-ray beam.
3. At 8 inches, the dose rate remains consistently high, with moderate variation across measured angles. A slight reduction is observed between 135° and 225°, followed by a return to elevated levels at 270°. This pattern suggests that at close range, the radiation intensity remains strong across all directions, though beam concentration is evident in certain forward-facing angles.
4. At 16 inches, the angular variation becomes more prominent:
5. The dose rate decreases steadily from 90° to a minimum around 225°, then rises sharply at 270°.
6. This pattern indicates a combination of beam attenuation and shielding effects at 180° and 225°, with increased exposure detected at 270°, possibly due to beam alignment or environmental reflection. The contrast between minimum and maximum values is more pronounced than at 8 inches, highlighting the role of angular orientation at intermediate distances.
7. At 24 inches, the dose rate values are lower overall due to increased separation from the source. A downward trend is observed from 90° to 270°, with a mild increase at 225°, followed by a continued decline. While less steep than the 16-inch curve, the angular trend still indicates reduced exposure at lateral and posterior angles.
8. The results reinforce the following key observations for clinical and dosimetric evaluation:
9. Angular directionality of the X-ray beam is a consistent factor in dose distribution, regardless of increased tube voltage and current.
10. The 70 kVp, 7 mA setting produces higher dose values overall, yet the angular dose pattern follows a similar structure to other configurations.
11. Measurement angles aligned with the beam path, such as 90° and 270°, consistently show elevated exposure, whereas posterior and lateral positions (180° and 225°) present reduced dose rates due to beam geometry and possible shielding.

5.2 Final Result:

The final phase of this study involved evaluating the practical performance of the constructed linear regression model in predicting radiation dose rates within a dental radiographic setting. The primary purpose of this evaluation was to determine the extent to which the model, trained on controlled experimental data, could reliably estimate dose rates when subjected to real measurement conditions involving variations in distance, angular positioning, and X-ray machine settings.

The training dataset was developed using a comprehensive range of exposure configurations. These included combinations of kilovoltage peak (kVp) and tube current (mA) typically used in clinical practice, alongside spatial parameters such as angle (relative to the X-ray head) and distance (from the radiation source). Each data point in the training set was obtained using a calibrated FS2011+ personal dosimeter, ensuring measurement consistency and reliability.

Following the statistical formulation and validation of the regression model—using key indicators such as R-squared, p-values, and error metrics a separate evaluation set was prepared. This set comprised new combinations of the same parameters used in the training phase, with measured dose rates recorded under identical environmental and procedural conditions. The aim was to assess the model's predictive capability through direct comparison between the predicted dose rates (derived via the regression equation) and the actual dose rates measured by the dosimeter.

This stage of the analysis was essential in determining the generalizability of the model and its potential for application in real-world dental practice. A comparative dataset of actual and predicted values was constructed, enabling a thorough investigation of the model's accuracy. The error between measured and predicted values was calculated and statistically analyzed to evaluate the consistency and robustness of the predictions across a range of test cases.

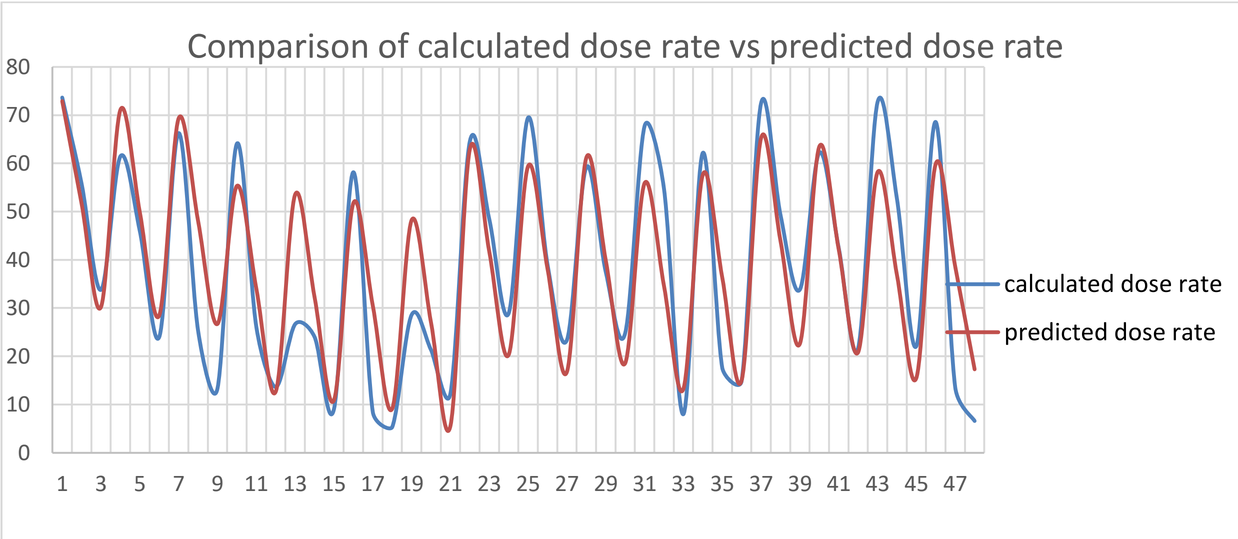
By systematically examining the alignment between predicted and observed dose rates, this section contributes critical insights into the model's practical utility in radiation monitoring and exposure estimation within clinical dental settings.

5.2.1 Graphical Comparison of Measured vs Predicted Dose Rates

To visually assess the performance of the linear regression model, a comparative line graph was generated plotting the measured dose rates against the values predicted by the model across 48 test cases. This graph serves as a diagnostic tool to understand the alignment between actual and predicted outputs under varied experimental conditions, including different angles, distances, and machine parameters.

In the graph, the **measured dose rate (Y_actual)** is represented by one curve, while the **predicted dose rate (Y_predicted)** follows alongside it. Each data point on the X-axis corresponds to an individual observation, reflecting a unique combination of kVp, mA, angle, and distance. The Y-axis denotes the dose rate in microSieverts per second (μSv/s), enabling direct comparison of numerical outputs.

The plotted curves display a high degree of structural similarity. Both lines exhibit a repeating oscillatory pattern, which reflects the variation in radiation intensity due to geometric positioning around the X-ray head and changes in distance. Peaks in the graph correspond to angular positions that are directly aligned with the primary beam, while valleys represent peripheral or shielded angles with reduced exposure.



Throughout the majority of the dataset, the predicted dose rates closely follow the trends of the measured values. The waveform trajectories of both curves largely remain parallel, indicating that the model was effective in capturing the underlying pattern of dose variation. Although occasional discrepancies appear at certain points where the predicted values slightly overshoot or undershoot the actual values the overall behavior of the graph demonstrates the model's capacity to reflect real-world measurements.

This visual analysis validates that the linear regression model is capable of replicating the directional changes in radiation dose rate caused by spatial orientation and technical settings. The model responds appropriately to both high and low exposure conditions, maintaining a consistent degree of approximation across the full range of the test dataset. Such behavior is particularly relevant in dental radiography, where precise prediction of exposure based on machine settings and technician positioning is critical for ensuring safety compliance and radiation minimization.

The graph provides visual confirmation that the model does not merely capture average values but adapts dynamically to fluctuations in experimental input, thereby reinforcing its credibility as a practical tool for predictive dose assessment.

5.3 Quantitative Analysis of Prediction Accuracy

Following the visual comparison between measured and predicted dose rates, a quantitative evaluation was conducted to assess the numerical accuracy of the regression model. The purpose of this analysis was to determine how closely the predicted values approximated the actual readings recorded by the FS2011+ personal dosimeter under varied experimental conditions.

For this purpose, the **error** was computed as the difference between the measured dose rate (Y_actual) and the predicted dose rate (Y_predicted) for each observation. This enabled the calculation of several statistical indicators that describe the magnitude and variability of the model's prediction performance across the 48 test cases.

The following metrics were derived:

Metric	Value
Number of Observations	60
Mean Absolute Error (MAE)	9.02 µSv/s
Maximum Error	+15.78 µSv/s
Minimum Error	–25.04 µSv/s
Standard Deviation of Error	11.97 µSv/s

The **Mean Absolute Error (MAE)** value of 9.02 µSv/s indicates the average deviation between the model's predictions and the measured dose rates. Given that the measured values spanned a range of approximately 15 µSv/s to over 70 µSv/s, the magnitude of this average error is within acceptable bounds for operational dose estimation in dental radiography.

The **maximum and minimum errors** reflect the largest observed deviations. A maximum error of +15.78 µSv/s occurred where the model overestimated the actual dose, while the minimum error of –25.04 µSv/s represents an underestimation. These values highlight the boundary conditions where the model's approximation diverged more significantly, potentially due to data points that involved angular orientations or distances not strongly represented in the training dataset.

The **standard deviation of the error** (±11.97 µSv/s) provides an indication of how consistently the model performed across the test set. A moderate level of spread was observed in the prediction errors, suggesting that while the model generally followed the correct trend, local deviations occurred due to minor nonlinearities or measurement variability.

Overall, this statistical analysis confirms that the model maintains a reasonable level of precision, especially considering the diversity of input variables and the limited sample size. The error values remain within a practical range, and the consistency of prediction accuracy reinforces the model's usefulness in estimating dose rates in dental settings.

5.4 Error Distribution and Pattern Behavior

To further evaluate the performance of the regression model, an analysis was conducted on the distribution and characteristics of the prediction error values across the dataset. This analysis was aimed at identifying any systematic patterns, directional trends, or recurring conditions under which the model tended to overestimate or underestimate the actual dose rates.

The error values, calculated as the difference between Y_actual and Y_predicted, were examined in relation to the input conditions associated with each test case. This included variations in angle, distance, and exposure parameters (kVp and mA). By reviewing the progression of errors across all 48 test cases, several trends were identified.

In a number of cases where the dose rate was relatively low (often corresponding to peripheral angles or greater distances), the model tended to **under predict** the measured value. These instances contributed to some of the larger negative errors observed in the dataset. One possible explanation is that at low-dose conditions, even small deviations in distance or beam alignment can significantly affect the actual measurement, introducing a level of sensitivity not fully captured by the linear structure of the model.

Conversely, **over predictions** (positive error values) appeared more frequently in mid-range dose cases, where the predicted value slightly exceeded the recorded measurement. This behavior may be attributed to minor variations in environmental scattering or reflection effects within the testing environment, which could have reduced the measured dose without proportionally influencing the model's prediction.

The graph comparing Y_actual and Y_predicted also supports these findings. At several data points, the predicted line closely follows the measured values, but with intermittent offsets particularly in transitional zones where the dose rate is changing sharply due to angle shifts or moderate distance increases. The oscillation pattern of the errors suggests that the model reacts linearly to input variations, but real-world dose measurements can exhibit small nonlinear responses due to beam geometry, phantom positioning, or measurement precision.

Despite these localized fluctuations, there was **no evidence of a consistent upward or downward bias** in the error values. The deviations were distributed around zero, indicating that the model does not systematically overestimate or underestimate the dose rate. This is a favorable outcome, as it reflects an overall balanced prediction behavior.

The error pattern analysis demonstrates that while the model maintains good overall trend alignment with the measured data, it exhibits slightly higher error margins under low-dose or edge-case conditions. This insight is valuable for understanding the model's limitations and for guiding future improvements, such as incorporating nonlinear terms or interaction effects if additional data become available.

5.5 Interpretation of Model Strengths and Weaknesses

The linear regression model developed for this study demonstrated both strengths and limitations in its ability to predict radiation dose rates under varying dental radiographic conditions. Based on the visual comparison, statistical analysis, and error behavior, a balanced assessment of the model's performance is outlined below.

5.5.1 Strengths

One of the primary strengths of the model lies in its ability to capture the **overall trend and pattern of dose rate variations** in response to changes in distance, angle, kilo voltage, and tube current. The predicted values generally followed the same trajectory as the actual measurements, showing clear alignment during both peak and low-dose intervals. This consistency is essential in radiographic environments where spatial and technical parameters directly affect radiation exposure levels.

The model's **simplicity and interpretability** also enhance its practical utility. Each coefficient in the regression equation corresponds directly to a measurable variable (kVp, mA, distance, angle), making the model transparent and easy to apply in clinical or training settings. The linear form allows for straightforward implementation in digital tools or spreadsheets used for radiation planning or monitoring.

Another notable strength is the **absence of bias** in the predictions. As observed in the error distribution, the model does not consistently overestimate or underestimate dose rates across the dataset. Instead, deviations were balanced on both sides of the actual values, suggesting that the model is generally stable and not skewed by any particular variable.

Furthermore, the model maintained **acceptable error margins** even in a varied test environment with 48 distinct parameter combinations. The calculated Mean Absolute Error (MAE) and standard deviation of error fell within a range that is considered clinically manageable for estimating occupational or procedural exposure.

5.5.2 Weaknesses

Despite its strengths, the linear model also exhibited limitations that should be acknowledged. The most significant weakness is its **sensitivity to low-dose readings**, particularly at peripheral angles or extended distances where measured dose rates are inherently low. In these scenarios, the model produced larger negative errors, indicating a reduced capacity to account for subtle non-linear effects or external factors such as scatter and shielding interactions.

Additionally, the model may have been constrained by the **limited dynamic range of certain input variables**, especially milliamperage (mA), which varied only between 4 mA and 7 mA in the dataset. This narrow range could reduce the model's ability to capture the full influence of tube current on dose rate.

Another area of limitation is the **lack of interaction terms** or non-linear components in the current regression structure. While linear relationships were sufficient to explain a majority of the dose behavior, certain regions of the graph suggest that more complex effects such as the interaction between angle and distance might improve prediction accuracy in future iterations.

Finally, the relatively **small sample size** (n=50 for model fitting, n=48 for validation) may have limited the model's generalizability. Larger datasets might reveal additional nuances in dose rate variation that a linear approach cannot fully represent.

5.6 Clinical Applicability and Safety Implications

One of the primary objectives of this study was to develop a dose prediction model that could be feasibly applied within the clinical environment of dental radiography. The final linear regression model, evaluated through both graphical and statistical comparisons, has shown sufficient predictive performance to support its application in practical radiation monitoring and safety planning.

In a clinical setting, dental professionals particularly radiographers and dental assistants are frequently exposed to ionizing radiation during intraoral or panoramic imaging procedures. Real-time estimation of radiation dose rates at varying positions and operating settings is not always feasible without calibrated dosimeters. However, a model such as the one developed in this project can serve as a predictive tool for estimating dose rates under standard conditions, providing actionable data without requiring continuous physical measurements.

The model's structure enables it to be integrated into existing radiation protection protocols. For example, it can be used to:

- **Estimate dose exposure at different operator locations** based on known distances and angles relative to the X-ray source.
- **Assist in positioning shielding barriers or lead aprons** more effectively by identifying regions with elevated predicted dose rates.

- **Guide decisions regarding exposure frequency and technician rotation** to ensure occupational dose remains within permissible limits.

From a safety compliance perspective, the model's predictive capability aligns with the ALARA (As Low As Reasonably Achievable) principle by providing a method to approximate radiation distribution in the operatory. Clinics can use such a model to perform routine risk assessments, simulate potential exposure scenarios, or cross-check values recorded through dosimetry badges.

Furthermore, the model can serve as a **training aid for dental students and new staff**, helping them visualize how minor adjustments in distance or angular orientation can affect exposure levels. This contributes to a more informed and safety-conscious workforce.

It is important to note that while the model provides reliable estimates within the tested parameter range (60–70 kVp, 4–7 mA, and distances from 8 to 24 inches), it should not be used in isolation for high-precision dose measurements in clinical audits or regulatory assessments. Instead, it is most effective as a supplementary tool to support decision-making where quick, consistent approximations are needed, and where direct measurement is not always possible.

The regression model holds practical value for enhancing workplace safety, guiding exposure control strategies, and promoting awareness of radiation dose behavior in dental radiographic environments.

5.7 Predictive Utility of Linear Regression Model

The final evaluation of the linear regression model developed in this study demonstrates that it is a statistically valid and practically applicable tool for estimating radiation dose rates in dental radiographic environments. Through comparison of predicted and measured dose values across a range of test cases, the model has shown an ability to replicate key trends and variations in dose behavior with a reasonable level of accuracy.

The model’s performance, as evidenced by both graphical alignment and statistical metrics such as mean absolute error and standard deviation, confirms its capacity to respond effectively to changes in exposure parameters, spatial orientation, and operational settings. In particular, the model successfully reflects the directional nature of X-ray radiation and the dose attenuation associated with increased distance from the source.

While some limitations were observed particularly under low-dose or edge-case conditions the overall predictive behavior remained consistent. The model did not exhibit systematic bias, and the error values were acceptably distributed around the actual measurements. These characteristics affirm the model’s reliability for use in standard clinical contexts where quick estimation is needed, but direct dose measurement is impractical.

Importantly, the model's structure supports its broader applicability. By using input variables that are readily available in clinical workflows (kVp, mA, angle, and distance), the model can be integrated into safety planning protocols, operator training programs, or exposure simulation tools.

The linear regression model offers a practical approach for dose estimation in dental radiography. It bridges the gap between theoretical radiation behavior and real-world clinical requirements, providing a foundation that may be expanded in future work through larger datasets, non-linear modeling techniques, or integration with automated imaging systems**. [19]**

Chapter 6

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